

Ch: 1. Ethics and Professionalism

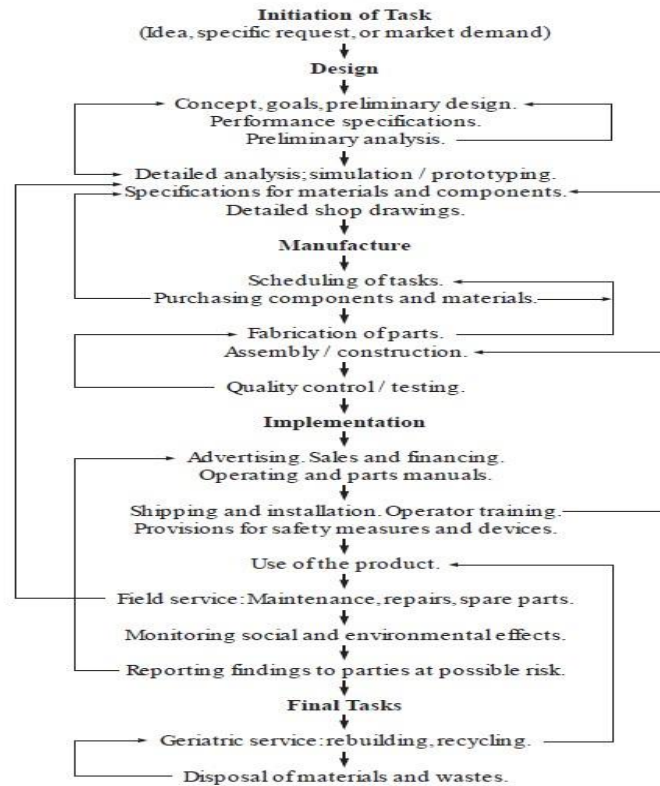
Ethics and Excellence in Engineering: Moral values are embedded in engineering projects as standards of excellence, not “tacked on” as external burdens. This is true of even the simplest engineering projects, as illustrated by the following assignment given to students in a freshman engineering course: “Design a chicken coop that would increase egg and chicken production, using materials that were readily available and maintainable by local workers [at a Mayan cooperative in Guatemala]. The end users were to be the women of a weaving cooperative who wanted to increase the protein in their children’s diet in ways that are consistent with their traditional diet, while not appreciably distracting from their weaving.”² The task proved more complex than first appeared. Students had to identify feasible building materials, decide between cages or one large fenced area, and design structures for strength and endurance. They had to create safe access for the villagers, including ample head and shoulder room at entrances and a safe floor for bare feet. They had to ensure humane conditions for the chickens, including adequate space and ventilation, comfort during climate changes, convenient delivery of food and water, and protection from local predators that could dig under fences. They also had to improve cleaning procedures to minimize damage to the environment while recycling chicken droppings as fertilizers. The primary goal, however, was to double current chicken and egg production. A number of design concepts were explored before a variation of a fenced-in concept proved preferable to a set of cages. Additional modifications needed to be made as students worked with villagers to implement the design in ways that best served their needs and interests. In combining myriad design goals and constraints, engineering projects integrate multiple moral values connected with those goals and constraints—for example, safety, efficiency, respect for persons, and respect for the environment. As elsewhere, moral values are myriad, and they can give rise to *ethical dilemmas*: situations in which moral reasons come into conflict, or in which the applications of moral values are problematic, and it is not immediately obvious what should be done. The moral reasons might be obligations, rights, goods, ideals, or other moral considerations. For example, at what point does the aim of increasing chicken and egg production compromise humane conditions for the animals? Technical skill and morally good judgment need to go together in solving ethical dilemmas, and, in general, in making moral choices. So do competence and conscientiousness, creativity and good character. These combinations were identified by the ancient Greeks, whose word *arete* translates into English as “excellence” or as “virtue.” In engineering, as in other professions, excellence and ethics go together—for the most part and in the long run.

ii. Mirco and Macro Issues. Today, engineers are increasingly asked to understand excellence and ethics in terms of broader societal and environmental concerns. They need to be prepared to grapple with both micro and macro issues. Micro issues concern the decisions made by individuals and companies in pursuing their projects. Macro issues concern more global issues, such as the directions in technological development, the laws that should or should not be passed, and the collective responsibilities of groups such as engineering professional societies and consumer groups. Both micro and macro issues are important in engineering ethics, and often they are interwoven.³ As an illustration, consider debates about sport utility vehicles (SUVs). Micro issues arose concerning the Ford Explorer and also Bridgestone/Firestone, who provided tires for the Explorer. During the late 1990s, reports began to multiply about the tread on Explorer tires separating from the rest of the tire, leading to blowouts and rollovers. By 2002, estimates were that 300 people had died, and another 1,000 people were injured, and more recent estimates place the numbers much higher. Ford and Bridgestone/ Firestone blamed each other for the problem, leading to the breakup of a century-old business partnership. As it turned out, the hazard had multiple sources.

Bridgestone/Firestone used a flawed tire design and poor quality control at a major manufacturing facility. Ford chose tires with a poor safety margin, relied on drivers to maintain proper inflation within a very narrow range, and then dragged its feet in admitting the problem and recalling dangerous tires. In contrast, macro issues center on charges that SUVs are among the most harmful vehicles on the road, especially given their numbers. The problems are many: gas-guzzling, excessive polluting, instability because their height leads to rollovers, greater “kill rate” of other drivers during accidents, reducing the vision of drivers in shorter cars behind them on freeways, and

blinding other drivers’ vision because of high-set lights. Keith Bradsher estimates that SUVs are causing approximately 3,000 deaths in excess of what cars would have caused: “Roughly 1,000 extra deaths occur each year in SUVs that roll over, compared with the expected rollover death rate if these motorists had been driving cars. About 1,000 more people die each year in cars hit by SUVs than would occur if the cars had been hit by other cars. And up to 1,000 additional people succumb each year to respiratory problems because of the extra smog caused by SUVs.”⁴ Bradsher believes these numbers will continue to increase as more SUVs are added to the road each year and as older vehicles are resold to younger and more dangerous drivers. Should “the SUV issue” be examined within engineering as a whole, or at least by representative professional and technical societies? If so, what should be done? Or, in a democratic and capitalistic society, should engineers play a role only as individuals but not as organized groups? Should engineers remain uninvolved, leaving the issue entirely to consumer groups and lawmakers? We leave these questions as discussion questions at the end of the section. Even larger macro issues surround public transportation issues in relation to all automobiles and SUVs as we look to the future with a dramatically increasing population, a shrinking of traditional resources, and concerns about global warming.

iii. Dimensions in Engineering: Let us gain a more detailed understanding of moral complexity in engineering as a product develops from a mental concept to physical completion. Engineers encounter both moral and technical problems concerning variability in the materials available to them, the quality of work by coworkers at all levels, pressures imposed by time and the whims of the marketplace, and relationships of authority within corporations. Figure 1–2 charts the sequence of tasks that leads from the concept of a product to its design, manufacture, sale, use, and ultimate disposal. The idea of a new product is first captured in a conceptual design, which will lead to establishing performance specifications and conducting a preliminary analysis based on the functional relationships among design variables. These activities lead to a more detailed analysis, possibly assisted by computer simulations and physical models or prototypes. The end product of the design task will be detailed specifications and shop drawings for all components. Manufacturing is the next major task. It involves scheduling and carrying out the tasks of purchasing materials and components, fabricating parts and subassemblies, and finally assembling and performancetesting the product. Selling comes next, or delivery if the product is the result of a prior contract. Thereafter, either the manufacturer’s or the customer’s engineers perform installation, personnel training, maintenance, repair, and ultimately recycling or disposal. Seldom is the process carried out in such a smooth, continuous fashion as indicated by the arrows progressing down the middle of Figure 1–2. Instead of this uninterrupted sequence, intermediate results during or at the end of each stage often require backtracking to make modifications in the design developed thus far. Errors need to be detected and corrected. Changes may be needed to improve product performance or to meet cost and time constraints. An altogether different, alternative design might have to be considered. In the words of Herbert Simon, “Design is usually the kind of problem solving we call ill-structured . . . you don’t start off with a well-defined goal. Nor do you start off with a clear set of alternatives, or perhaps any alternatives at



all. Goals and alternatives have to emerge through the design process itself: One of its first tasks is to clarify goals and to begin to generate alternatives.”⁵ This results in an iterative process, with some of the possible recursive steps indicated by the thin lines and arrows on either side of Figure 1–2. As shown, engineers are usually forced to stop during an initial attempt at a solution when they hit a snag or think of a better approach. They will then return to an earlier stage with changes in mind. Such reconsiderations of earlier tasks do not necessarily start and end at the same respective stages during subsequent passes through design, manufacture, and implementation. That is because the retracing is governed by the latest findings from current experiments, tempered by the outcome of earlier iterations and experience with similar product designs. Changes made during one stage will not only affect subsequent stages but might also require a reassessment of prior decisions. Dealing with this complexity requires close cooperation among the engineers of many different departments and disciplines such as chemical, civil, electrical, industrial, and mechanical engineering. It is not uncommon for engineering organizations to suffer from “silo mentality,” which makes engineers disregard or denigrate the work carried out by groups other than their own. It can be difficult to improve a design or even to rectify mistakes under such circumstances. Engineers do well to establish contact with colleagues across such artificial boundaries so that information can be exchanged more freely. Such contacts become especially important in tackling morally complex problems.

iv. Potential Moral Problems. To repeat, engineering generally does not consist of completing designs or processes one after another in a straightforward progression of isolated tasks. Instead, it involves a trial-and-error process with backtracking based on decisions made after examining results obtained along the way. The design iterations resemble feedback loops, and like any well-functioning feedback control system, engineering takes into account natural and social environments that affect the product and people using it.⁶ Let us therefore revisit the engineering tasks, this time as listed in Table 1–1, along with examples of problems that might arise.

The grab bag of problems in Table 1–1 can arise from shortcomings on the part of engineers, their supervisors, vendors, or the operators of the product. The underlying causes can have different forms:

1. *Lack of vision*, which in the form of tunnel vision biased toward traditional pursuits overlooks suitable alternatives, and in the form of groupthink promotes acceptance at the expense of critical thinking.
2. *Incompetence* among engineers carrying out technical tasks.
3. *Lack of time or lack of proper materials*, both ascribable to poor management.
4. A *silo mentality* that keeps information compartmentalized rather than shared across different departments.
5. The notion that there are safety engineers *somewhere down the line* to catch potential problems.
6. *Improper use or disposal of the product* by an unwary owner or user.
7. *Dishonesty* in any activity shown in Figure 1–2 and pressure by management to take shortcuts.
8. *Inattention* to how the product is performing after it is sold and when in use.

Although this list is not complete, it hints at the range of problems that can generate moral challenges for engineers. It also suggests why engineers need foresight and caution, especially in imagining who might be affected indirectly by their products and by their decisions, in good or harmful ways.

v. Why Study Engineering Ethics?

Engineering ethics should be studied because it is *important*, both in contributing to safe and useful technological products and in giving meaning to engineers' endeavors. It is also *complex*, in ways that call for serious reflection throughout a career, beginning with earning a degree. But beyond these general observations, what specific aims should guide the study of engineering ethics?

In our view, the direct aim is to increase our ability to deal effectively with moral complexity in engineering. Accordingly, the study of engineering ethics strengthens our ability to reason clearly and carefully about moral questions. To invoke terms widely used in ethics, the unifying goal is to increase moral autonomy. Autonomy means *self-determining*, but not just any kind of independent reflection about ethics amounts to moral autonomy. Moral autonomy can be viewed as the skill and habit of thinking rationally about ethical issues on the basis of moral concern and commitment. This foundation of general responsiveness to moral values derives primarily from the training we receive as children in being sensitive to the needs and rights of others, as well as of ourselves. When such training is absent, as it often is with seriously abused children, the tragic result can be an adult sociopath who lacks any sense of moral right and wrong. Sociopaths (or

psychopaths) are not morally autonomous, regardless of how independent their intellectual reasoning about ethics might be. Improving the ability to reflect carefully on moral issues can be accomplished by improving various practical skills that will help produce autonomous thought about moral issues. As related to engineering ethics, these skills include the following.

1. Moral awareness: Proficiency in recognizing moral problems and issues in engineering
2. Cogent moral reasoning: Comprehending, clarifying, and assessing arguments on opposing sides of moral issues
3. Moral coherence: Forming consistent and comprehensive viewpoints based on consideration of relevant facts
4. Moral imagination: Discerning alternative responses to moral issues and finding creative solutions for practical difficulties
5. Moral communication: Precision in the use of a common ethical language, a skill needed to express and support one's moral views adequately to others These are the *direct* goals in college courses. They center on cognitive skills—skills of the intellect in thinking clearly and cogently. It is possible, however, to have these skills and yet not act in morally responsible ways. Should we therefore add to our list of goals the following goals that specify aspects of moral commitment and responsible conduct? 6. Moral reasonableness: The willingness and ability to be morally reasonable
7. Respect for persons: Genuine concern for the well-being of others as well as oneself
8. Tolerance of diversity: Within a broad range, respect for ethnic and religious differences and acceptance of reasonable differences in moral perspectives
9. Moral hope: Enriched appreciation of the possibilities of using rational dialogue in resolving moral conflicts
10. Integrity: Maintaining moral integrity and integrating one's professional life and personal convictions

In our view we should add these goals to the study of engineering ethics, for without them there would be little practical point in studying ethics. At the same time, the goals are often best pursued implicitly and indirectly, more in how material is studied and taught than in preaching and testing. A foundation of moral concern must be presupposed, as well as evoked and expanded, in studying ethics at the college level.

vi. All Cases in the 'Discussion Questions' part of the book.

1. Identify the moral values, issues, and dilemmas, if any, involved in the following cases, and explain why you consider them moral values and dilemmas.

a. An engineer notified his firm that for a relatively minor cost a flashlight could be made to last several years longer by using a more reliable bulb. The firm decides that it would be in its interests not to use the new bulb, both to keep costs lower and to have the added advantage of “built-in obsolescence” so that consumers would need to purchase new flashlights more often.

b. A linear electron accelerator for therapeutic use was built as a dual-mode system that could either produce X-rays or electron beams. It had been in successful use for some time, but every now and then some patients received high overdoses, resulting in painful after-effects and several deaths. One patient on a repeat visit experienced great pain, but the remotely located operator was unaware of any problem because of lack of communication between them: The intercom was broken, and the video monitor had been unplugged. There also was no way for the patient to exit the examination chamber without help from the outside, and hence the hospital was partly at fault. On cursory examination of the machine, the manufacturer insisted that the computerized and automatic control system could not possibly have malfunctioned and that no one should spread unproven and potentially libelous information about the design. It was the

painstaking, day-and-night effort of the hospital's physicist that finally traced the problem to a software error introduced by the manufacturer's efforts to make the machine more user-friendly.

2. Regarding the following example, comment on why you think simple human contact made such a large difference. What does it say about what motivated the engineers, both before and after the encounter? Is the case too unique to permit generalizations to other engineering products? A team of engineers are redesigning an artificial lung marketed by their company. They are working in a highly competitive market, with long hours and high stress. The engineers have little or no contact with the firm's customers, and they are focused on technical problems, not people. It occurs to the project engineer to invite recipients of artificial lungs and their families to the plant to talk about how their lives were affected by the artificial lung. The change is immediate and striking: "When families began to bring in their children who for the first time could breathe freely, relax, learn, and enjoy life because of the firm's product, it came as a revelation. The workers were energized by concrete evidence that their efforts really did improve people's lives, and the morale of the workplace was given a great lift."

3. Should SUV problems at the macro level be of concern to engineers as a group and their professional societies? Should individual automotive engineers, in their daily work, be concerned about the general social and environmental impacts of SUVs?

4. It is not easy to define morality in a simple way, but it does not follow that morality is a hopelessly vague notion. For a long time, philosophers thought that an adequate definition of any idea would specify a set of logically necessary and sufficient conditions for applying the idea. For example, each of the following features is logically necessary for a triangle, and together they are sufficient: a plane figure, having three straight lines, closed to form three angles. The philosopher Ludwig Wittgenstein (1889–1951), however, argued that most ordinary (nontechnical) ideas cannot be neatly defined in this way. Instead, there are often only "family resemblances" among the things to which words are applied, analogous to the partly overlapping similarities among members of a family—similar eye color, shape of nose, body build, temperament, and so forth.¹⁰ Thus, a book might be hardback, paperback, or electronic; printed or handwritten; in English or German; and so forth. Can you specify necessary and sufficient conditions for the following ideas: chairs, buildings, energy, safety, engineers, morality?

5. Mention of ethics sometimes evokes groans, rather than engagement, because it brings to mind onerous constraints and unpleasant disagreements. Worse, it evokes images of self-righteousness, hypocrisy, and excessively punitive attitudes of blame and punishment—attitudes that are themselves subject to moral critique. Think of a recent event that led to a public outcry. With regard to the event, discuss the difference between being morally reasonable and being moralistic in a pejorative sense. In doing so, consider such things as breadth of vision, tolerance, sensitivity to context, and commitment.

Ch: 2. Moral Reasoning and Codes of Ethics

Moral Choices and Ethical Dilemmas.

Designing Aluminum Cans

Henry Petroski chronicles the development of aluminum beverage cans with stay-on tab openers.¹ Aluminum cans are now ubiquitous—approximately 100 billion are produced in the United States each year. The first aluminum can was designed in 1958 by Kaiser Aluminum, in the attempt to improve on heavier and more expensive tin cans. Aluminum proved ideal as a lightweight, flexible material that allowed manufacturing of the bottom and sides of the can from a single sheet, leaving the top to be added after the can was filled. The trick was to make the can strong enough to keep the pressurized liquid inside, while being thin enough to be cost-effective. The can also had to fit conveniently in the hand and reliably satisfy customers' needs. Design calculations solved the problem of suitable thickness of material, but improvements came gradually in shaping of the inward-dished bottom to improve stability when the can is set down, as well as to provide some leeway for expansion of the can. The first aluminum cans, like the tin cans before them, were opened with a separate opener, which required additional manufacturing costs to make them readily available to consumers. The need for separate openers also caused inconvenience, as Ermal Frazee discovered when, forgetting an opener while on a picnic in 1959, he had to resort to using a car bumper. Frazee, who owned Dayton Reliable Tool and Manufacturing Company and was hence familiar with metal, envisioned a design for a small lever that was attached to the can but which was removed as the can opened. The idea proved workable and was quickly embraced by manufacturers. Gradual improvements were made over subsequent years to ensure easy opening and prevention of lip and nose injuries from the jagged edges of the opening. Within a decade an unanticipated crisis arose, however, creating an ethical dilemma. Frazee had not thought through the implications of billions of discarded pull tabs causing pollution, foot injuries, and harm to fish and infants who ingested them. The dilemma was what to do to balance usefulness to consumers with protection of the environment. A technological innovation solved the dilemma in a manner that integrated all the relevant values. In 1976 Daniel F. Cudzick invented a simple, stay-attached opener of the sort familiar today. Once again, minor design improvements came as problems were identified. Indeed, the search for improvements continues today because people with arthritic fingers or long and breakable fingernails have difficulty using the current openers. All the while, of course, the broader problem of pollution from cans themselves prompted recycling programs that now recycle more than six out of ten cans (leaving room for further improvement here as well).

Petroski recounts these developments to illustrate how engineering progresses by learning from design failures—that is, designs that cause unacceptable risks or other problems. At each stage of the design process, engineers are preoccupied with what might go wrong. The hope is to anticipate and prevent failures, drawing on knowledge about past failures. Here, however, our interest is in how moral values were embedded in the design dilemmas concerning the environment.

If we understand moral choices broadly, as decisions involving moral values, then the development of aluminum cans can be understood as a series of routine moral choices interspersed with occasional moral dilemmas. Moral values entered implicitly into the decision-making process of engineers and their managers—decisions that probably appeared to be purely technical or purely economic. This appearance is misleading, for the technical and economic decisions had moral dimensions in four general directions:

safety, environmental protection, consumer usefulness, and economic benefits. First, human safety is obviously a moral value, rooted directly in the moral worth of human beings. Some aspects of safety seem minor—slight cuts to lips and noses from poorly designed openers and minor injuries to feet in recreation areas such as beaches. But minor injuries might cause infections, and even by themselves they have some moral significance. Again, various kinds of poisoning might occur unless all materials were tested under a range of conditions, and there are potential industrial accidents during the manufacturing process. Finally, extensive testing was needed to ensure that exploding cans, although not inherently dangerous, did not cause automobile accidents when drivers were distracted while opening cans. A second set of moral values concern the environment. Many of these values overlap with the first set, safety. Billions of detached can openers raised the level of hazards to people walking with bare feet. Injuries to fish and other wildlife posed additional concerns. Depending on one's environmental ethic, injuries to wildlife might be understood as direct moral harms to creatures recognized as having inherent worth, or instead as indirect harms to human beings. The broader problem of environmental pollution from aluminum cans and their openers required corporate action in paying for recycled materials, community action in developing the technologies for recycling, and changes in public policy and social attitudes about recycling. Third, some moral values are masked under terms such as *useful* and *convenient* products. We tend to think of such matters as nonmoral, especially with regard to trivial things such as sipping a carbonated beverage with a pleasing taste. But there are moral connections, however indirect or minor. After all, water is a basic need, and convenient access to pleasant-tasting liquids contributes to human well-being. However slightly, these pleasures bear on human happiness and well-being, especially when considered on the scale of massproduced products. In addition, the aesthetic values pertaining to the shape and appearance of cans have some relevance to satisfying human desires.

Finally, the economic benefits to stakeholders in the corporation have moral implications. Money matters, and it matters morally. Jobs provide the livelihood for workers and their families that make possible the material goods that contribute to happiness—and survival. The corporation's success contributes as well to the livelihood of suppliers and retailers, as well as to stockholders. All these values—safety, environmental protection, convenience, and money—were relevant throughout the development of aluminum cans, not merely when they explicitly entered into moral dilemmas. Hence, the case illustrates how moral values permeate engineering practice.

ii. Steps in Resolving Ethical Dilemmas:

Reasonable solutions to ethical dilemmas are clear, informed, and well-reasoned. *Clear* refers to moral clarity—clarity about which moral values are at stake and how they pertain to the situation. It also refers to conceptual clarity—precision in using the key concepts (ideas) applicable in the situation. *Informed* means knowing and appreciating the implications of morally-relevant facts. In addition, it means being aware of alternative courses of action and what they entail. *Well-reasoned* means that good judgment is exercised in integrating the relevant moral values and facts to arrive at a morally desirable solution. These characteristics of reasonable solutions also enter as steps in resolving ethical dilemmas. By “steps” we do not mean singlefile movements, but instead activities that are carried out jointly and in repeating patterns. Thus, a preliminary survey of the applicable moral values and relevant facts might be followed by conceptual clarification and additional fact gathering, which in turn evince a more nuanced understanding of the applicable values and the implications of the relevant facts. In discussing the example, we will illustrate the importance of professional codes of ethics in identifying and highlighting applicable moral reasons. A chemical engineer working in the environmental division of a computer manufacturing firm learns that her company might be discharging unlawful amounts of lead and arsenic into the city sewer.² The city processes the sludge into a fertilizer used by local farmers. To ensure the safety of both the discharge and the fertilizer, the city imposes restrictive laws on the discharge of lead and arsenic. Preliminary investigations convince the engineer that the company should implement stronger pollution controls but her supervisor tells her the

cost of doing so is prohibitive and that technically the company is in compliance with the law. She is also scheduled to appear before town officials to testify in the matter. What should she do?

1. Moral clarity: Identify the relevant moral values. The most basic step in confronting ethical dilemmas is to become aware of them! This means identifying the moral values and reasons applicable in the situation, and bearing them in mind as further investigations are made. These values and reasons might be obligations, rights, goods, ideals (which might be desirable but not mandatory), or other moral considerations. Exactly how we articulate the relevant values reflects our moral outlook. Hence, the moral frameworks discussed in Chapter 3 are relevant even in stating what the ethical dilemma is. Another resource is talking with colleagues, who can help sharpen our thinking about what is at stake in the situation. But the most useful resource in identifying ethical dilemmas in engineering are professional codes of ethics, as interpreted in light of one's ongoing professional experience. Like most codes of ethics, the code of ethics of the American Institute of Chemical Engineers (AIChE) indicates the engineer has at least three responsibilities in the situation. One responsibility is to be honest: "Issue statements or present information only in an objective and truthful manner." A second responsibility is to the employer: "Act in professional matters for each employer or client as faithful agents or trustees, avoiding conflicts of interest and never breaching confidentiality." A third responsibility is to the public, and also to protect the environment: "Hold paramount the safety, health, and welfare of the public and protect the environment in performance of their professional duties." In the case at hand, the members of the public most directly affected are the local farmers, but the dangerous chemicals could affect more persons as lead and arsenic are drawn into the food chain. Additional moral considerations, not cited in the code, include duties to maintain personal and professional integrity, and rights to pursue one's career.

2. Conceptual clarity: Be clear about key concepts. Professionalism requires being a faithful agent of one's employer, but does that mean doing what one's supervisor directs or doing what is good for the corporation in the long run? These might be different things, in particular when one's supervisor is adopting a shortterm view that could harm the long-term interests of the corporation. Again, what does it mean to "hold paramount the safety, health, and welfare of the public" in the case at hand? Does it pertain to all threats to public health, or just serious threats, and what is a "serious" threat? Again, does being "objective and truthful" simply mean never lying (intentionally stating a falsehood), or does it mean revealing all pertinent facts (withholding nothing important) and doing so in a way that gives no preference to the interests of one's employer over the needs of the public to be informed of hazards?

3. Informed about the facts: Obtain relevant information. This means gathering information that is pertinent in light of the applicable moral values (as identified in step 1). Sometimes the primary difficulty in resolving moral dilemmas is uncertainty about the facts, rather than conflicting values per se. Certainly in the case at hand, the chemical engineer needs to check and recheck her findings, perhaps asking colleagues for their perspectives. Her corporation seems to be violating the law, but is it actually doing so? We, like the engineer, need to know more about the possible harm caused by the minute quantities of lead and arsenic over time. How serious is it, and how likely to cause harm?

4. Informed about the options: Consider all (realistic) options. Initially, ethical dilemmas seem to force us into a two-way choice: Do this or do that. Either bow to a supervisor's orders or blow the whistle to the town authorities. A closer look often reveals additional options. (Sometimes writing down the main options and suboptions as a matrix or decision tree ensures that all options are considered.) The chemical engineer might be able to suggest a new course of research that will improve the removal of lead and arsenic. Or she might discover that the city's laws are needlessly restrictive and should be revised. Perhaps she can think of a way to convince her supervisor to be more open-minded about the situation, especially given the possible damage to the corporation's image if it should later be found in violation of the law. Unless an emergency develops, these and other steps should be attempted before informing authorities outside the corporation—a desperate last resort, especially given the likely penalties for whistle-blowing (see Chapter 7).

5. Well-reasoned: *Make a reasonable decision.* Arrive at a carefully reasoned judgment by weighing all the relevant moral reasons and facts. This is not a mechanical process that a computer or algorithm might do for us. Instead, it is a deliberation aimed at integrating all the relevant reasons, facts, and values—in a morally reasonable manner. If there is no ideal solution, as is often the case, we seek a satisfactory one, what Herbert Simon dubbed “satisficing.”

Often a code of ethics provides a straightforward solution to dilemmas, but not always. Codes are not recipe books that together with precise hierarchies of relative stringency among the rules. What about the case at hand? The code does assert one very important hierarchy: *Hold paramount the public safety, health, and welfare.* The AIChE code also requires engineers to “formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public.” This statement, combined with the statement of the paramount responsibility, makes it clear that the responsibility to be a faithful agent of the employer does not override professional judgment in important matters of public safety. At the same time, the recommendation to “consider further disclosure, if warranted” seems somewhat lukewarm, both because it is placed parenthetically and because it only says “consider.” It suggests something to think about, rather than a firm statement of duty. As such, it is weaker than statements in some other codes, including the code of the National Society of Professional Engineers (NSPE), that require notification of appropriate authorities when one’s judgment is overridden in matters where public safety is endangered. Which of these codes takes precedence?

Furthermore, exactly what does the paramount statement entail in the case at hand? If the engineer is convinced her company produces valuable computers, might she reasonably conclude that the public good is held paramount by coming “close enough” to obeying the law? As for the requirement to be “objective and truthful,” that certainly implies not lying to the townofficials, but might she reasonably conclude she is being objective by not divulging information her supervisor says is confidential? Obviously, such conclusions might be products of rationalization (biased reasoning), rather than sound moral reasoning. We mention them only to suggest that codes are no substitute for morally good judgment—honest, fair, responsible moral judgment. Indeed, as we have just seen, good judgment is needed even in interpreting the code of ethics.³ The development of good moral judgment is part and parcel of developing experience in engineering. It is also a primary goal in studying ethics.

iii. Codes of Ethics - Importance of Codes.

A code of ethics is a set of principles and guidelines that outline the moral and professional standards of conduct for individuals or organizations in a particular profession, industry, or community. It serves as a framework for ethical decision-making and guides individuals in their interactions and responsibilities.

Codes of ethics state the moral responsibilities of engineers as seen by the profession and as represented by a professional society. Because they express the profession’s collective commitment to ethics, codes are enormously important, not only in stressing engineers’ responsibilities but also in supporting the freedom needed to meet them. Codes of ethics play at least eight essential roles: serving and protecting the public, providing guidance, offering inspiration, establishing shared standards, supporting responsible professionals, contributing to education, deterring wrongdoing, and strengthening a profession’s image.

1. Serving and protecting the public. Engineering involves advanced expertise that professionals have and the public lacks, and also considerable dangers to a vulnerable public. Accordingly, professionals stand in a fiduciary relationship with the public: Trust and trustworthiness are essential. A code of ethics functions as a commitment by the profession as a whole that engineers will serve the public health, safety, and welfare. In one way or another, the remaining functions of codes all contribute to this primary function. **2. Guidance.** Codes provide helpful guidance by articulating the main obligations of engineers. Because codes should be brief to be effective, they offer mostly general guidance. Nonetheless, when well written, they identify

primary responsibilities. More specific directions may be given in supplementary statements or guidelines, which tell how to apply the code.

3. Inspiration. Because codes express a profession's collective commitment to ethics, they provide a positive stimulus (motivation) for ethical conduct. In a powerful way, they voice what it means to be a member of a profession committed to responsible conduct in promoting the safety, health, and welfare of the public. Although this paramount ideal is somewhat vague, it expresses a collective commitment to the public good that inspires individuals to have similar aspirations.

4. Shared standards. The diversity of moral viewpoints among individual engineers makes it essential that professions establish explicit standards, in particular minimum (but hopefully high) standards. In this way, the public is assured of a standard of excellence on which it can depend, and professionals are provided a fair playing field in competing for clients.

5. Support for responsible professionals. Codes give positive support to professionals seeking to act ethically. A publicly proclaimed code allows an engineer, under pressure to act unethically, to say: "I am bound by the code of ethics of my profession, which states that . . ." This by itself gives engineers some group backing in taking stands on moral issues. Moreover, codes can potentially serve as legal support for engineers criticized for living up to work-related professional obligations. **6. Education and mutual understanding.** Codes can be used by

professional societies and in the classroom to prompt discussion and reflection on moral issues. Widely circulated and officially approved by professional societies, codes encourage a shared understanding among professionals, the public, and government organizations about the moral responsibilities of engineers. A case in point is NSPE's BER, which actively promotes moral discussion by applying the NSPE code to cases for educational purposes.

7. Deterrence and discipline. Codes can also serve as the formal basis for investigating unethical conduct. Where such investigation is possible, a deterrent for immoral behavior is thereby provided. Such an investigation generally requires paralegal proceedings designed to get at the truth about a given charge without violating the personal rights of those being investigated. Unlike the American Bar Association and some other professional groups, engineering societies cannot by themselves revoke the right to practice engineering in the United States. Yet some professional societies do suspend or expel members whose professional conduct has been proven unethical, and this alone can be a powerful sanction when combined with the loss of respect from colleagues and the local community that such action is bound to produce.

8. Contributing to the profession's image. Codes can present a positive image to the public of an ethically committed profession. Where warranted, the image can help engineers more effectively serve the public. It can also win greater powers of self-regulation for the profession itself, while lessening the demand for more government regulation. The reputation of a profession, like the reputation of an individual professional or a corporation, is essential in sustaining the trust of the public.

iv. **Abuse of Codes.**

When codes are not taken seriously within a profession, they amount to a kind of window dressing that ultimately increases public cynicism about the profession. Worse, codes occasionally stifle dissent within the profession and are abused in other ways. Probably the worst abuse of engineering codes is to restrict honest moral effort on the part of individual engineers to preserve the profession's public image and protect the status quo. Preoccupation with keeping a shiny public image may silence healthy dialogue and criticism. And an excessive interest in protecting the status quo may lead to a distrust of the engineering profession on the part of both government and the public. The best way to increase trust is by encouraging and helping engineers to speak freely and responsibly about public safety and well-being. This includes a tolerance for criticisms of the codes themselves, rather than allowing codes to become sacred documents that have to be accepted uncritically. On rare occasions, abuses have discouraged moral conduct and caused serious harm

to those seeking to serve the public. For example, two engineers were expelled from American Society of Civil Engineers (ASCE) for violating a section of its code forbidding public remarks critical of other engineers. Yet the actions of those engineers were essential in uncovering a major bribery scandal related to the construction of a dam for Los Angeles County. Moreover, codes have sometimes placed unwarranted “restraints of commerce” on business dealings to benefit those within the profession. Obviously there is disagreement about which, if any, entries function in these ways. Consider the following entry in the pre-1979 versions of the NSPE code: The engineer “shall not solicit or submit engineering proposals on the basis of competitive bidding.” This prohibition was felt by the NSPE to best protect the public safety by discouraging cheap engineering proposals that might slight safety costs to win a contract. The Supreme Court ruled, however, that it mostly served the self-interest of established engineering firms and actually hurt the public by preventing the lower prices that might result from greater competition (*National Society of Professional Engineers v. United States* [1978]).

v. Ethical Relativism.

Does a profession’s code of ethics create the obligations that are incumbent on members of the profession, so that engineers’ obligations are entirely relative to their code of ethics? Or does the code simply record the obligations that already exist? One view is that codes try to put into words obligations that already exist, whether or not the code is written. As Stephen Unger writes, codes “recognize” obligations that already exist: “A code of professional ethics may be thought of as a collective recognition of the responsibilities of the individual practitioners”; codes cannot be “used in cookbook fashion to resolve complex problems,” but instead they are “valuable in outlining the factors to be considered.”¹⁴ Unger takes codes very seriously as profession’s shared voice in articulating the responsibilities of its practitioners. A good code provides valuable focus and direction, but it does not generate obligations so much as articulate obligations that already exist. Michael Davis disagrees, and he places far greater emphasis on professional codes of ethics. In his view, codes are conventions established within professions to promote the public good. As such, they are morally authoritative. The code itself generates obligations: “a code of ethics is, as such, not merely good advice or a statement of aspiration. It is a standard of conduct which, if generally realized in the practice of a profession, imposes a *moral* obligation on each member of the profession to act accordingly.” Notice the word “imposes,” as distinct from “recognizing” an obligation that already exists. To violate the code is wrong because it creates an unfair advantage in competing with other professionals in the marketplace. Davis has been accused of endorsing *ethical relativism*, also called ethical conventionalism, which says that moral values are entirely relative to and reducible to customs—to the conventions, laws, and norms of the group to which one belongs.¹⁶ What is right is simply what conforms to custom, and it is right solely *because* it conforms to customs. We can never say an act is objectively right or obligatory without qualification, but only that it is right for members of a given group because it is required by their customs. In the words of anthropologist Ruth Benedict, “We recognize that morality differs in every society, and is a convenient term for socially approved habits. Mankind has always preferred to say, ‘It is morally good,’ rather than ‘It is habitual.’ . . . But historically the two phrases are synonymous.”¹⁷ In particular, professional ethics is simply the set of conventions embraced by members of a profession, as expressed in their code. There are problems with ethical relativism, whether we are talking about the conventions of a profession such as engineering or the conventions of a society in its entirety. By viewing customs as selfcertifying, ethical relativism rules out the possibility of critiquing the customs from a wider moral framework. For example, it leaves us without a basis for criticizing genocide, the oppression of women and minorities, child abuse, torture, and reckless disregard of the environment, when these things are the customs of another culture. Regarding professional ethics, ethical relativism implies that we cannot morally critique a given code of ethics, giving reasons for why it is justified in certain ways and perhaps

open to improvement in other ways. Ethical relativism also seems to allow any group of individuals to form its own society with its own conventions, perhaps ones that common sense tells us are immoral. Again, an engineer might be a member of one or more professional societies, a weapons development corporation and a pacifist religious tradition, and the customs of these groups in matters of military work might point in different directions. Although ethical relativism is a dubious moral outlook, it remains true that moral judgments are made “in relation to” particular circumstances, such as those of engineering. It is also true that customs are “morally relevant”(though not always decisive) in deciding how we ought to behave. Finally, *some* moral requirements are indeed established by mutual agreements. Just as laws establish the legal and moral permissibility of driving on the right side of the road (in the United States) or the left side (in England), some requirements in engineering codes of ethics create obligations. For example, some of the specific conflicts of interest forbidden in codes of ethics are forbidden by agreement within the profession to ensure fair competition among engineers.

In our view, then, Unger and Davis are both partly correct. Unger is correct in holding that many of the entries in codes of ethics state responsibilities that would exist regardless of the code—for example, to protect the safety, health, and welfare of the public. Davis is correct that some parts of codes are conventions arrived at by mutual agreement within the profession.

Ch: 3. Moral Frameworks

i. Right Ethics, Duty Ethics, Utilitarianism

Rights are moral entitlements and valid moral claims that impose duties on other people. All ethical theories leave some room for rights, but the ethical theory called rights ethics is distinctive in that it makes human rights the ultimate appeal—the moral bottom line. Human rights constitute a moral authority to make legitimate moral demands on others to respect our choices, recognizing that others can make similar claims on us. At its core, rights ethics emphasizes respecting the inherent dignity and worth of individuals as they exercise their liberty. Rights ethics is the most familiar ethical theory, for it provides the moral foundation of the political and legal system of the United States. Thus, in the Declaration of Independence Thomas Jefferson wrote: “We hold these truths to be self-evident; that all men are created equal; that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty, and the pursuit of Happiness.” Unalienable—or inalienable, natural, human—rights cannot be taken away (alienated) from us, although of course they are sometimes violated. Human rights have been appealed to in all the major social movements of the twentieth century, including the women’s movement, the civil rights movement, the farm workers’ movement, the gay rights movement, and the patients’ rights movement (in health care). The idea of human rights is the single most powerful moral concept in making cross-cultural moral judgments about customs and laws.¹ As such, the notions of human rights and legal rights are distinct. Legal rights are simply those the law of a given society says one has, whereas human rights are those we have as humans, whether the law recognizes them or not. Rights ethics applies to engineering in many ways. It provides a powerful foundation for the special ethical requirements in engineering and other professions.² Most engineering codes of ethics enjoin holding paramount the safety, health, and welfare of the public, a requirement that can be interpreted as having respect for the public’s rights to life, rights not to be injured by dangerous products, rights to privacy, and rights to receive benefits through fair and honest exchanges in a free marketplace. In addition, the basic right to liberty implies a right to give informed consent to the risks accompanying technological products, an idea developed in Chapter 4. In addition to human rights, there are special moral rights—rights held by particular individuals rather than by every human being. For example, engineers and their employers have special moral rights that arise from their respective roles and

the contracts they make with each other. Special rights are grounded in human rights, however indirectly. Thus, contracts and other types of promises create special rights because people have human rights to liberty that are violated when the understandings and commitments specified in contracts and promises are violated. And when consumers purchase products, there is an implicit contract, based on an implicit understanding, that the products will be safe and useful. Rights ethics gets more complex as we ask which kinds of rights exist—only liberty rights, or also welfare rights? Liberty rights are rights to exercise our liberty, and they place duties on other people not to interfere with our freedom. (The “not” explains why they are also called negative rights.) Welfare rights are rights to benefits needed for a decent human life, when we cannot earn those benefits, perhaps because we are severely handicapped, and when the community has them available. (As a contrast to negative rights, they are sometimes called positive rights.) Most rights ethicists affirm that both liberty and welfare human rights exist.³ Indeed, they contend that liberty rights imply at least some basic welfare rights. What, after all, is the point of saying that we have rights to liberty if we are utterly incapable of exercising liberty because, for example, we are unable to obtain the basic necessities, such as jobs, worker compensation for serious injuries, and health care? Shifting to legal rights, most Americans also support selective welfare rights, including a guaranteed public education of kindergarten through twelfth grade, Medicare and Medicaid, Social Security, and reasonable accommodations for persons with disabilities. Another influential version of rights ethics, however, denies there are welfare human rights. Libertarians believe that only liberty rights exist; there are no welfare rights. John Locke (1632–1704), who was the first philosopher to carefully articulate a rights ethics, is often interpreted as a libertarian.⁴ Locke’s version of human rights ethics was highly individualistic. He viewed rights primarily as entitlements that prevent other people from meddling in our lives. The individualistic aspect of Locke’s thought is reflected in the contemporary political scene in the Libertarian political party and outlook, with its emphases on protecting private property, dismantling welfare systems, and opposition to extensive government regulation of business and the professions. Libertarians take a harsh view of taxes and government involvement beyond the bare minimum necessary for national defense, a legal system, and the preservation of free enterprise. Locke thought of property as whatever we gained by “mixing our labor” with things—for example, coming to own lumber by going into the wilderness and cutting down a tree. Today, however, our understanding of property is far more complex. Many believe that property is largely what the law and government specify as to how we can acquire and use material things. Even so, Locke’s followers tended to insist that property was sacrosanct and that governments continually intruded on property rights, particularly in the form of excessive taxation and regulation. Finally, both libertarians and other rights ethicists can agree that few rights are absolute, in the sense of being unlimited and having no justifiable exceptions. When rights conflict with rights in practical situations, thereby creating ethical dilemmas, good judgment is required in arriving at reasonable solutions about how to reasonably balance the rights.

DUTY ETHICS:

Rights and duties are typically correlated with each other. For example, our right to life places duties on others not to kill us, and our right to liberty places duties on others not to interfere with our freedom. Duty ethics reverses the order of priority by beginning with duties and deriving rights from them. Although the similarities between duty ethics and rights ethics are pronounced, historically they developed as distinct moral traditions. Duty ethics says that right actions are those required by duties to respect the liberty or autonomy (self-determination) of individuals. One duty ethicist suggests the following list of important duties: “(1) Do not kill. (2) Do not cause pain. (3) Do not disable. (4) Do not deprive of freedom. (5) Do not deprive of pleasure. (6) Do not deceive. (7) Keep your promises. (8) Do not cheat. (9) Obey the law. (10) Do your duty [referring to work, family, and other special responsibilities].”⁵ How do we know that these are our duties? Immanuel Kant (1724–1804), the most famous duty ethicist, argued that all such specific duties derive from one fundamental duty to respect persons.⁶ Persons deserve respect because they are moral agents—capable of recognizing and voluntarily responding to moral duty (or, like children, they

potentially have such capacities). Autonomy—moral self-determination or self-governance—means having the capacity to govern one’s life in accordance with moral duties. Hence, respect for persons amounts to respect for their moral autonomy. We ought always to treat persons as having their own rational aims, and not merely use them for our ends. Immorality occurs when we reduce other people to mere means to our ends and needs. Violent acts such as murder, rape, and torture are obvious ways of treating people as mere objects serving our own purposes. We also fail to respect persons if we fail to provide support for them when they are in desperate need, and we can help them at little inconvenience to ourselves. Some duties, then, are to refrain from interfering with a person’s liberty, and some express requirements to help them when they are in need, thereby paralleling the distinction between liberty and positive rights. Of course we need to use one another as means all the time: Business partners, managers and engineers, and faculty and students use each other to obtain their personal and professional ends. Immorality occurs when we merely use persons as means to our goals, rather than as autonomous agents who have their own goals. We also have duties to ourselves, for we too are rational and autonomous beings. As examples, Kant said we have a duty not to commit suicide, which would bring an end to a valuable life; we have duties to develop our talents, as part of unfolding our rational natures; and we should avoid harmful drugs that undermine our ability to exercise our rationality. Obviously, Kant’s repeated appeal to the idea of rationality makes a number of assumptions about morally worthy aims. After beginning with the minimal idea of rationality as the capacity to obey moral principles, he built in a host of specific goals as part of what it means to be rational. Kant emphasized that duties are universal: They apply equally to all persons. Here again, the idea is that valid principles of duty apply to all rationally autonomous beings, and hence valid duties will be such that we can envision everyone acting on them. This idea of universal principles is often compared to the Golden Rule: Do unto others as you would have them do unto you; or, in its negative version, Do not do unto others what you would not want them to do to you.⁷ Finally, Kant insisted that moral duties are “categorical imperatives.” As imperatives, they are injunctions or commands that we impose on ourselves as well as other rational beings. As categorical, they require us to do what is right because it is right, unconditionally and without special incentives attached. For example, we should be honest because honesty is required by duty; it is required by our basic duty to respect the autonomy of others, rather than to deceive and exploit them for our own selfish purposes. “Be honest!” says morality—not because doing so benefits us, but because honesty is our duty. Morality is not an “iffy” matter that concerns hypothetical (conditional) imperatives, such as “If you want to prosper, be honest.” A businessperson who is honest solely because honesty pays—in terms of profits from customers who return and recommend their services, as well as from avoiding jail for dishonesty—fails to fully meet the requirements of morality. In this way, morality involves attention to motives and intentions, an idea also important in virtue ethics. Kant’s ideas of respect for autonomy, duties to ourselves, universal duties, and categorical imperatives have been highly influential. However, he made one large mistake. He thought that everyday principles of duty, such as “Do not lie” and “Keep your promises,” were absolute in the sense of never having justifiable exceptions. In doing so, he conflated three ideas: (1) universality—moral rules apply to all rational agents; (2) categorical imperatives—moral rules command what is right because it is right; and (3) absolutism—moral rules have no justified exceptions. Nearly all ethicists reject Kant’s absolutism, even ethicists who embrace his ideas of universality and categorical imperatives. The problem with absolutism should be obvious. As we emphasized in Chapter 2, moral reasons are many and varied, including those expressed by principles of duty. Given the complexity of human life, they invariably come into conflict with each other, thereby creating moral dilemmas. Contemporary duty ethicists recognize that many moral dilemmas are resolvable only by recognizing some valid exceptions to simple principles of duty. Thus, engineers have a duty to maintain confidentiality about information owned by their corporations, but that duty can be overridden by the paramount duty to protect the safety, health, and welfare of the public. To emphasize that most duties have some justified exceptions, philosophers now use the expression *prima facie* duties.⁸ In this technical sense, *prima facie* simply means “might have justified exceptions” (rather than “at first glance”). Most duties are *prima facie*—they sometimes have permissible or obligatory exceptions. Indeed, the same is true

of most rights and other moral principles, and hence today the term *prima facie* is also applied to rights and moral rules of all kinds.

Utilitarianism: Rights ethics and duty ethics agree that some types of actions, for example being fair and truthful, are (*prima facie*) obligatory for reasons independent of their consequences. In contrast, utilitarianism says the sole standard of right action is good consequences. There is only one general moral requirement: Produce the most good for the most people, giving equal consideration to everyone affected. (The word utility is sometimes used to refer to good consequences and other times to the balance of good over bad consequences.) At first glance, the utilitarian standard seems simple and plausible. Surely morality involves producing good consequences— especially in engineering. Indeed, utilitarian modes of thinking are reflected in cost-benefit analyses: Tally up the likely good consequences of various options or proposals; do likewise for the likely bad consequences; and then favor that proposal which maximizes the overall good. Utilitarianism also seems a straightforward way to interpret the central principle in most engineering codes: “Engineers shall hold paramount the safety, health and welfare of the public in the performance of their professional duties.” After all, welfare is a rough synonym for overall good (utility), and safety and health might be viewed as especially important aspects of that good. Yet, what exactly is the good to be maximized? And should we maximize the good with respect to each situation, or instead with regard to general rules (policies, laws, principles in codes of ethics)? Depending on how these questions are answered, utilitarianism takes different forms. To begin with, what is the standard for measuring good consequences? Specifically, what is intrinsic good—that is, good considered just by itself? All other good things are instrumental goods in that they provide means (instruments) for gaining intrinsic goods. Some utilitarians consider pleasure to be the only intrinsic good. But that seems counterintuitive—there is nothing good about the pleasures of tyrants and sadists take in inflicting suffering. John Stuart Mill believed that happiness was the only intrinsic good, and hence he understood utilitarianism as the requirement to produce the greatest amount of happiness.⁹ What is happiness? Sometimes Mill confused it with pleasures and enjoyments, which are short-term, feel-good states of consciousness. In the main, however, Mill thought of happiness as (a) a life rich in pleasures, especially the “higher” pleasures of friendship, love, and intellectual endeavors, mixed with some inevitable pains, plus (b) a pattern of activities and relationships that we can affirm as the way we want our lives to be. There are alternative theories of intrinsic good. Some utilitarians understand intrinsic goods as those which a reasonable person would pursue, or those which satisfy rational desires—those that we can affirm after fully examining them in light of relevant information, for example, love, friendship, appreciation of beauty, in addition to happiness. In sharp contrast, most economists adopt a preference theory: What is good is what individuals prefer, as manifested in what they choose to purchase in the marketplace. Arguments over which, if any, of these theories of intrinsic good obviously complicate utilitarian ethical theories. In addition to developing a plausible view of intrinsic good, we need to decide whether to focus on individual actions or general rules. Classical, nineteenth-century utilitarians such as Mill believed in act-utilitarianism: A particular action is right if it is likely to produce the most good for the most people in a given situation, compared with alternative options available. The standard can be applied at any moment. Right now, should you continue reading this chapter? You might instead take a break, go to sleep, see a movie, or pursue any number of other options. Each option would have both immediate and long-term consequences that can be estimated. The right action is the one that produces the most overall good, taking into account everyone affected. Yet, act-utilitarianism seems vulnerable to objections. It apparently permits some actions that we know, on other grounds, are patently immoral. Suppose that stealing a computer from my employer, an old one scheduled for replacement anyway, benefits me significantly and causes only miniscule harm to the employer and others. We know that the theft is unethical, and hence act-utilitarianism seems to justify wrongdoing. Again, suppose that in a particular situation more good is promoted by keeping the public ignorant about serious dangers, for example, by not informing them about a hidden fault in a car or building. Or suppose that it will improve company morale if several disliked engineers are fired after being blamed for mistakes they did not make. Doing so is unfair,

but the overall good is promoted. Such difficulties lead many, perhaps most, utilitarians to shift to an alternative version of utilitarianism that says we should maximize the good through following rules that maximize good consequences, rather than through isolated actions. According to this view, called ruleutilitarianism, right actions are those required by rules that produce the most good for the most people. Because rules interact with each other, we need to consider a set of rules. Thus, we should seek to discover and act on an optimal moral code—that set of rules which maximizes the public good more than alternative codes would (or at least as much as alternatives).¹⁰ Rule-utilitarians have in mind society-wide rules, but the same idea applies to rules stated in engineering codes of ethics. Thus, an engineering code of ethics is justified in terms of its overall good consequences (compared to alternative codes), and so engineers should abide by it even when an exception might happen to be beneficial. For example, if codified rules forbidding bribes and deception are justified, then even if a particular bribe or deception is beneficial in some situations, one should still refrain from them.

Virtue ethics:

Character is the pattern of virtues (morally desirable features) and vices (morally undesirable features) in persons. Virtues are desirable habits or tendencies in action, commitment, motive, attitude, emotion, ways of reasoning, and ways of relating to others. Vices are morally undesirable habits or tendencies. The words virtue and vice sound a bit old-fashioned. Words for specific virtues, however, remain familiar, both in engineering and in everyday life—for example, competence, honesty, courage, fairness, loyalty, and humility. Words for specific vices are also familiar: incompetence, dishonesty, cowardice, unfairness, disloyalty, and arrogance. Aristotle (384–322 BC) suggested that the moral virtues are habits of reaching a proper balance between extremes, whether in conduct, emotion, or desire.¹⁴ Virtues are tendencies to find the reasonable (golden) mean between the extremes of too much (excess) and too little (deficiency) with regard to particular aspects of our lives. Thus, truthfulness is the appropriate middle ground (mean) between revealing all information, in violation of tact and confidentiality (excess), and being secretive or lacking in candor (deficiency) in dealing with truth. Again, courage is the mean between foolhardiness (the excess of rashness) and cowardice (the deficiency of self-control) in confronting dangers. The most important virtue is practical wisdom, that is, morally good judgment, which enables us to discern the mean for all the other virtues. The Greek word arete translates as either “virtue” or “excellence,” an etymological fact that reinforces our theme of ethics and excellence going together in engineering. The most comprehensive virtue of engineers is responsible professionalism. This umbrella virtue implies four (overlapping) categories of virtues: public well-being, professional competence, cooperative practices, and personal integrity. Publicspirited virtues are focused on the good of clients and the wider public. The minimum virtue is nonmaleficence, that is, the tendency not to harm others intentionally. As Hippocrates reportedly said in connection with medicine, “Above all, do no harm.” Engineering codes of professional conduct also call for beneficence, which is preventing or removing harm to others and, more positively, promoting the public safety, health, and welfare. Also important is a sense of community, manifested in faith and hope in the prospects for meaningful life within professional and public communities. Generosity, which means going beyond the minimum requirements in helping others, is shown by engineers who voluntarily give their time, talent, and money to their professional societies and local communities. Finally, justice within corporations, government, and economic practices is an essential virtue in the profession of engineering. Proficiency virtues are the virtues of mastery of one’s profession, in particular mastery of the technical skills that characterize good engineering practice. Following Aristotle, some thinkers regard these values as intellectual virtues rather than distinctly moral ones. As they contribute to sound engineering, however, they are morally desirable features. The most general proficiency virtue is competence: being well prepared for the jobs one undertakes. Another is diligence: alertness to dangers and careful attention to detail in performing tasks by, for example, avoiding the deficiency of laziness and the excess of the workaholic. Creativity is especially desirable within a rapidly changing technological society. Teamwork virtues are those that are especially important in enabling professionals to work successfully with other people. They

include collegiality, cooperativeness, loyalty, and respect for legitimate authority. Also important are leadership qualities that play key roles within authority-structured corporations, such as the responsible exercise of authority and the ability to motivate others to meet valuable goals. Finally, self-governance virtues are those necessary in exercising moral responsibility.¹⁵ Some of them center on moral understanding and perception: for example, self-understanding and good moral judgment—what Aristotle calls practical wisdom. Other self-governance virtues center on commitment and on putting understanding into action: for example, courage, self-discipline, perseverance, conscientiousness, fidelity to commitments, self-respect, and integrity. Honesty falls into both groups of self-governance virtues, for it implies both truthfulness in speech and belief and trustworthiness in commitments. Virtue ethics takes contrasting forms, depending on which virtues are emphasized. To illustrate, we will contrast Samuel Florman's emphasis on conscientiousness and team-work virtues with Alasdair MacIntyre's emphasis on wider loyalties to community. Samuel Florman is famous for his celebration of the existential pleasures of engineering, that is, the deeply rooted and elemental satisfactions in engineering that contribute to happiness.¹⁶ These pleasures have many sources. There is the desire to improve the world, which engages an individual's sense of personal involvement and power. There is the challenge of practical and creative effort, including planning, designing, testing, producing, selling, constructing, and maintaining, all of which bring pride in achieving excellence in the technical aspects of one's work. There is the desire to understand the world—an understanding that brings wonder, peace, and sense of being at home in the universe. There is the sheer magnitude of natural phenomena—oceans, rivers, mountains, and prairies—that both inspires and challenges the design of immense ships, bridges, tunnels, communication links, and other vast undertakings. There is the presence of machines that can generate a comforting and absorbing sense of a manageable, controlled, and ordered world. Finally, engineers live with a sense of helping, of contributing to the well-being of other human beings. In elaborating on these pleasures, Florman implicitly sets forth a virtue ethics. In his view, the essence of engineering ethics is captured by the word conscientiousness, which combines competence and loyalty.¹⁷ "Competence" does not mean minimally adequate, but instead performing with requisite skill and experience. It implies exercising due care, persistence and diligence, honesty, attention to detail, and sometimes creativity. Loyalty means serving the interests of organizations that employ us, within the bounds of the law. Unlike libertarians who favor minimum government regulation, Florman places great emphasis on laws as setting the basic rules governing engineering. Within a democratic setting in which laws express a public consensus, economic competition among corporations makes possible technological achievements that benefit the public. Competition depends on engineers who are loyal to their organizations, rather than engineers "filtering their everyday work through a sieve of ethical sensitivity."¹⁸ Whereas Florman defends the priority of duties to employers, most professional codes that require engineers to hold paramount the safety, health, and welfare of the public, which does seem to imply "filtering their everyday work through a sieve of ethical sensitivity." Such an emphasis on the good of community is found in Alasdair MacIntyre's emphasis on public-spirited virtues in thinking about professions.¹⁹ MacIntyre conceives of professions as valuable social activities, which he calls social practices. A social practice is "any coherent and complex form of socially established cooperative human activity through which goods internal to that form of activity are realized in the course of trying to achieve those standards of excellence which are appropriate to, and partially definitive of, that form of activity."²⁰ To clarify, internal goods are good things (products, activities, experiences, etc.) that are so essential to a social practice that they partly define it. In engineering these goods are safe and useful technological products—products that can be further specified with regard to each area of engineering. Additional internal goods are personal goods connected with meaningful work, such as personal meaning in working as an engineer to create useful and safe public goods and services. In contrast, external goods can be earned in or outside specific professions, such as money, power, self-esteem, and prestige. Although both internal and external goods are important, excessive concern for external goods, whether by individuals or organizations, threatens internal goods and undermines social practices. Whether in Florman's or MacIntyre's version, virtue ethics seems vulnerable to the criticisms that it is incomplete and too vague. The meaning and

requirements of virtues need to be spelled out in terms of at least rough guidelines or rules, lest the virtues fail to provide adequate moral guidance.²¹ For example, honesty requires certain kinds of actions, done from certain kinds of motives. It implies a disposition, among other things, not to tell lies (without special justification) because lying disrespects persons and otherwise causes harm.

Self-realization: Each of the preceding ethical theories leaves considerable room for self-interest, that is, for pursuing what is good for oneself. Thus, utilitarianism says that self-interest should enter into our calculations of the overall good; rights ethics says we have rights to pursue our legitimate interests; duty ethics says we have duties to ourselves; and virtue ethics links our personal good with participating in communities and social practices. Self-realization ethics, however, gives greater prominence to self-interest and to personal commitments that individuals develop in pursuing self-fulfillment. As with the other ethical theories, we will consider two versions, this time depending on how the self (the person) is conceived. In a community-oriented version, the self to be realized is understood in terms of caring relationships and communities. In a second version, called ethical egoism, the self is conceived in a highly individualistic manner. The community-oriented version of self-realization ethics says that each individual ought to pursue self-realization, but it emphasizes the importance of caring relationships and communities in understanding self-realization. It emphasizes that we are social beings whose identities and meaning are linked to the communities in which we participate. This theme is expressed by F. H. Bradley (1826–1924): “The ‘individual’ apart from the community is an abstraction. It is not anything real, and hence not anything that we can realize. . . . I am myself by sharing with others.”²² Individuals vary greatly in what they desire most strongly, and also in their talents and virtues.²³ Self-realization ethics points to the highly personal commitments that motivate, guide, and give meaning to the work of engineers and other professionals.

These commitments enter into the core of an individual’s character.²⁴ As such, they reflect what engineers care about deeply in ways that evoke their interest and energy, shape their identities, and generate pride or shame in their work. Personal commitments are commitments that are not incumbent on everyone—for example, specific humanitarian, environmental, religious, political, aesthetic, supererogatory, and family commitments. They also include, however, commitments to obligatory professional standards, especially when these are linked to an individual’s broader value perspective. Personal commitments are relevant in many ways to professional life, including one’s choice of career and choice of jobs. Most important, they create meaning; thereby they motivate professionalism throughout long careers. Professions offer special opportunities for meaningful work, which explains much of their attraction to talented individuals. The relevant idea of meaning has subjective aspects—a “sense of meaning” that enlivens one’s daily work and life. It also has objective aspects—the justified values that make work worthwhile and help make life worth living. In the following passage Joanne B. Ciulla has in mind both subjective and objective meaning:

“Meaningful work, like a meaningful life, is morally worthy work undertaken in a morally worthy organization. Work has meaning because there is some good in it. The most meaningful jobs are those in which people directly help others or create products that make life better for people. Work makes life better if it helps others; alleviates suffering; eliminates difficult, dangerous, or tedious toil; makes someone healthier and happier; or aesthetically or intellectually enriches people and improves the environment in which we live.”²⁵ As just one illustration of personal commitments, and of the motivation and guidance they generate, consider the commitment to be creative, as illustrated by Jack Kilby who coined the microchip. The invention has had momentous importance in making possible the development of today’s powerful computers, so much so that in 2000 Kilby was awarded a Nobel Prize—a rare event for an engineer, as Nobel Prizes are usually given for fundamental contributions to science, not engineering. In retrospect, the idea behind the microchip seems simple, as do many creative breakthroughs. During the 1950s the miniaturization of transistors was being pursued at a relentless pace, but it was clear there would soon be a limit to the vast number of minute components that could be wired together. Kilby was well aware of the problem, sensed the need for a fundamentally new approach, and was driven by personal commitments to be creative in finding a solution. In July 1958, only a few weeks after starting a new job at

Texas Instruments, he discovered the solution: Make all parts of the circuit out of one material integrated on a piece of silicon, thereby removing the need to wire together miniature components. In making his discovery, Kilby was not pursuing a grand intention to provide humanity with the remarkable goods the microchip would make possible, although it is true he was known for his everyday kindness to colleagues. When he was about to give his Nobel lecture, he was introduced as having made the invention that “launched the global digital revolution, making possible calculators, computers, digital cameras, pacemakers, the Internet, etc.”²⁶ In response, he told a story borrowed from another Nobel laureate: “When I hear that kind of thing, it reminds me of what the beaver told the rabbit as they stood at the base of Hoover Dam: ‘No, I didn’t build it myself, but it’s based on an idea of mine.’ ” Was Kilby merely seeking money, power, fame, and other rewards just for himself? No, although these things mattered to him. As one biographer suggests, “we see nothing extraordinary in Jack Kilby’s private ambition or in his aim to find personal fulfillment through professional achievement. In that regard he was the same as the rest of us: We all pick professions with a mind to fulfilling ourselves.”²⁷ Primarily, Kilby was pursuing interests he had developed years earlier in how to solve technical problems in engineering. In this regard he was exceptional only in his passion for engineering work. Like many creative individuals, he was persistent to the point of being driven, and he found great joy in making discoveries. But even saying this by itself would be misleading. The accurate observation is that he had multiple motives, including motives to advance technology, to be compensated for his work, and to do some good for others. Building on this observation, we might sort the motives of professionals into three categories: proficiency, compensation, and moral. Proficiency motives, and their associated values, center on excellence in meeting the technical standards of a profession, together with related aesthetic values of beauty. The undergraduate curriculum for engineering is generally acknowledged to be more rigorous and difficult than the majority of academic disciplines. We might guess that students are attracted to engineering in part because of the challenge it offers to intelligent people. Do empirical studies back up this somewhat flattering portrayal? To a significant extent, yes. Typically, students are motivated to enter engineering primarily by a desire for interesting and challenging work. They have an “activist orientation” in the sense of wanting to create concrete objects and systems—to build them and to make them work. They are more skilled in math than average college students, although they tend to have a low tolerance for ambiguities and uncertainties that cannot be measured and translated into figures.²⁸ Compensation motives are for social rewards such as income, power, recognition, and job or career stability. We tend to think of these motives and values as self-interested, and in a large degree they are. Yet most people seek money for additional reasons, such as to benefit family members or even to be able to help others in need. In addition, financial independence prevents one from becoming a burden on others. In general, due regard for one’s self-interest is a moral virtue—the virtue of prudence—assuming it does not crowd out other virtues. Moral motives include desires to meet one’s responsibilities, respect the rights of others, and contribute to the well-being of others. Such motives of moral respect and caring involve affirming that other people have inherent moral worth. In addition, moral concern involves maintaining self-respect and integrity—valuing oneself as having equal moral worth and seeking to develop one’s talents. For the most part, these motives are interwoven and mutually supportive. All of them, not only moral motives, contribute to providing valuable services to the community, as well as professional relationships among engineers, other involved workers, and clients. Engineering is demanding, and it requires engineers to summon and to integrate a wide range of motivations. Indeed, life itself is demanding, and it can be argued that our survival requires constant interweaving and cross-fertilization of motives.²⁹ For many engineers, we should add, moral motivation and commitments are interwoven with spiritual and religious ones. Here are two examples. Egbert Schuurman is a Dutch Calvinist engineer who has written extensively on technology. Highlighting the dangers of technology, he calls for redirecting technology to serve morally worthy aims, both human liberation and respect for the environment. He and his coauthors of *Responsible Technology* articulate normative principles for design. They include: cultural appropriateness (preserving valuable institutions and practices within a particular society); openness (divulging to the public the value judgments expressed in products and also their known effects);

stewardship (frugality in the use of natural resources and energy); harmony (effectiveness of products together with promoting social unity); justice (respect for persons); caring (for colleagues and workers); and trustworthiness (deserving consumers' trust).³⁰ The second example is Mark Pesce, who invented dialup networking. In 1994, Pesce and a colleague developed the Virtual Reality Modeling Language (VRML), which allowed three dimensional models to be placed on the World Wide Web. Emphasizing the importance of spiritual attitudes in his work, he makes it clear that his beliefs are neither orthodox nor associated solely with any one world religion. He characterizes his beliefs as "a mélange of a lot of different religious traditions, including Christian, pre-Christian, Buddhist, Taoist and so on," integrated into a type of "paganism," which is "a practice of harmony, a religion of harmony with yourself and the environment."³¹ He is aware that his contributions to technology can be used as tools of communication or weapons of domination. Spiritual attitudes seek ways to allow aspects of the sacred into technology, to find ways for technology to make human life more interconnected through global communication, as well as attuned to nature, and to allow individuals to express themselves in more broadly creative ways through the Web. These examples barely hint at the myriad ways in which personal commitments, ideals, and meaning enter into professional ethics, including how individuals construe codified responsibilities.³² Later we offer additional illustrations; for example, in Chapter 7 we comment on personal commitments in connection with whistle-blowing, and in Chapter 8 we comment on environmental commitments.

Ch: 5. Commitment to Safety:

i. Safety and Risk: We demand safe products and services, but we also realize that we may have to pay for this safety. To complicate matters, what may be safe enough for one person may not be for someone else—either because of different perceptions about what is safe or because of different predispositions to harm. A power saw in the hands of a child will never be as safe as it can be in the hands of an adult. And, a sick adult is more prone to suffer ill effects from air pollution than is a healthy adult. Absolute safety, in the senses of (a) entirely risk-free activities and products, or (b) a degree of safety that satisfies all individuals or groups under all conditions, is neither attainable nor affordable. Yet it is important that we come to some understanding of what we mean by safety.

The concept of safety: One approach to defining safety would be to render the notion thoroughly subjective by defining it in terms of whatever risks a person judges to be acceptable. Such a definition was given by William W. Lowrance: “A thing is safe if its risks are judged to be acceptable.”² This approach helps underscore the notion that judgments about safety are tacitly value judgments about what is acceptable risk to a given person or group. Differences in appraisals of safety are thus correctly seen as reflecting differences in values. Lowrance’s definition, however, needs to be modified, for it departs too far from our common understanding of safety. This can be shown if we consider three types of situations. Imagine, first, a case where we seriously underestimate the risks of something, say of using a toaster we see at a garage sale. On the basis of that mistaken view, we judge it to be very safe and buy it. On taking it home and trying to make toast with it, however, it sends us to the hospital with a severe electric shock or burn. Using the ordinary notion of safety, we conclude we were wrong in our earlier judgment: The toaster was not safe at all. Given our values and our needs, its risks should not have been judged acceptable earlier. Yet, by Lowrance’s definition, we would be forced to say that prior to the accident the toaster was entirely safe because, after all, at that time we had judged the risks to be acceptable. Consider, second, the case where we grossly overestimate the risks of something. For example, we irrationally think fluoride in drinking water will kill a fifth of the populace. According to Lowrance’s definition, the fluoridated water is unsafe, because we judge its risks to be unacceptable. Third, there is the situation in which a group makes no judgment at all about whether the risks of a thing are acceptable or not—they simply do not think about it. By Lowrance’s definition, this means the thing is neither safe nor unsafe with respect to that group. Yet this goes against our ordinary ways of thinking about safety. For example, we normally say that some cars are safe and others unsafe, even though many people may never even think about the safety of the cars they drive. There must be at least some objective point of reference outside ourselves that allows us to decide whether our judgments about safety are correct once we have settled on what constitutes to us an acceptable risk. An expanded definition could capture this element, without omitting the insight already noted that safety judgments are relative to people’s value perspectives.³ One option is simply to equate safety with the absence of risk. Because little in life, and nothing in engineering, is risk-free, we prefer to adopt a modified version of Lowrance’s definition: A thing is safe if, were its risks fully known, those risks would be judged acceptable by reasonable persons in light of their settled value principles. In our view, then, safety is a matter of how people would find risks acceptable or unacceptable if they knew the risks and were basing their judgments on their most settled value perspectives. To this extent safety is an objective matter. It is a subjective matter to the extent that value perspectives differ. In what follows we will usually speak of safety simply as acceptable risk. But this is merely for convenience, and it should be interpreted as an endorsement of Lowrance’s definition only as we have qualified it. Safety is often thought of in terms of degrees and comparisons. We speak of something as “fairly safe” or “relatively safe” (compared with similar things). Using our definition, this translates as the degree to which a person or group, judging on the basis of their settled values, would decide that the risks of something are more or less acceptable in comparison with the risks of some other thing, and in light of relevant information. For example, when we say that airplane travel is safer than automobile travel, we mean that for each mile traveled it leads to fewer deaths and injuries—

the risky elements that our settled values lead us to avoid. Finally, we interpret “things” to include products, services, institutional processes, and disaster protection.

Risk: We say a thing is not safe if it exposes us to unacceptable risk; but what is meant by “risk”? A risk is the potential that something unwanted and harmful may occur. We take a risk when we undertake something or use a product or substance that is not safe. William D. Rowe refers to the “potential for the realization of unwanted consequences from impending events.”⁴ Thus a future, possible occurrence of harm is postulated. Risk, like harm, is a broad concept covering many different types of unwanted occurrences. In regard to technology, it can equally well include dangers of bodily harm, of economic loss, or of environmental degradation. These in turn can be caused by delayed job completion, faulty products or systems, or economically or environmentally injurious solutions to technological problems. Good engineering practice has always been concerned with safety. But as technology’s influence on society has grown, so has public concern about technological risks increased. In addition to measurable and identifiable hazards arising from the use of consumer products and from production processes in factories, some of the less obvious effects of technology are now also making their way to public consciousness. Although the latter are often referred to as new risks, many of them have existed for some time. They are new only in the sense that (1) they are now identifiable—because of changes in the magnitude of the risks they present, because they have passed a certain threshold of accumulation in our environment, or because of a change in measuring techniques, or (2) the public’s perception of them has changed—because of education, experience, media attention, or a reduction in other hitherto dominant and masking risks. Meanwhile, natural hazards continue to threaten human populations. Technology has greatly reduced the scope of some of these, such as floods, but at the same time it has increased our vulnerability to other natural hazards, such as earthquakes, as they affect our ever-greater concentrations of population and cause greater damage to our finely tuned technological networks of long lifelines for water, energy, and food. Of equal concern are our disposal services (sewers, landfills, recovery and neutralizing of toxic wastes) and public notification of potential hazards they present.

Ch: 6. Workplace Responsibilities and Rights

Confidentiality: The duty of confidentiality is the duty to keep secret all information deemed desirable to keep secret. Deemed by whom? Basically, it is any information that the employer or client would like to have kept secret to compete effectively against business rivals. Often this is understood to be any data concerning the company's business or technical processes that are not already public knowledge. Although this criterion is somewhat vague, it clearly points to the employer or client as the main source of the decision as to what information is to be treated as confidential. "Keep secret" is a relational expression. It always makes sense to ask, "Secret with respect to whom?" In the case of some government organizations, such as the Federal Bureau of Investigation (FBI) and Central Intelligence Agency (CIA), highly elaborate systems for classifying information have been developed that identify which individuals and groups may have access to what information. Within other governmental agencies and private companies, engineers and other employees are usually expected to withhold information labeled "confidential" from unauthorized people both inside and outside the organization. Several related terms should be distinguished. Privileged information literally means "available only on the basis of special privilege," such as the privilege accorded an employee working on a special assignment. Proprietary information is information that a company owns or is the proprietor of, and hence is a term carefully defined by property law. A rough synonym for "proprietary information" is trade secret, which can be virtually any type of information that has not become public, which an employer has taken steps to keep secret, and which is thereby given limited legal protection in common law (law generated by previous court rulings) that forbids employees from divulging it. Patents legally protect specific products from being manufactured and sold by competitors without the express permission of the patent holder. Trade secrets have no such protection, and a corporation can learn about a competitor's trade secrets through legal means—for instance, "reverse engineering," in which an unknown design or process can be traced out by analyzing the final product. But patents do have the drawback of being public and thus allowing competitors an easy means of working around them by finding alternative designs.

Confidentiality and Changing Jobs: The obligation to protect confidential information does not cease when employees change jobs. If it did, it would be impossible to protect such information. Former employees would quickly divulge it to their new employers or, perhaps for a price, sell it to competitors of their former employers. Thus, the relationship of trust between employer and employee in regard to confidentiality continues beyond the formal period of employment. Unless the employer gives consent, former employees are barred indefinitely from revealing trade secrets. This provides a clear illustration of the way in which the professional integrity of engineers involves much more than mere loyalty to one's present employer. Yet thorny problems arise in this area. Many engineers value professional advancement more than long-term ties with any one company, and so they change jobs frequently. Engineers in research and development are especially likely to have high rates of turnover. They are also the people most likely to be exposed to important new trade secrets. Moreover, when they transfer into new companies they often do the same kind of work as before—precisely the type of situation in which trade secrets of their old companies may have relevance, a fact that could have strongly contributed to their having readily found new employment. A high-profile case of trade secret violations was settled in January 1997 (without coming to trial) when Volkswagen AG (VW) agreed to pay General Motors Corporation (GM) and its German subsidiary Adam Opel \$100 million in cash and to buy \$1 billion in parts from GM over the next seven years. Why? Because in March 1993, Jose Ignacio Lopez, GM's highly effective manufacturing expert, left GM to join VW, a fierce competitor in Europe, and took with him not only three colleagues and knowhow, but also copies of confidential GM documents. A more legally important case concerned Donald Wohlgenuth, a chemical engineer who at one time was manager of B.F. Goodrich's space suit division.³ Technology for space suits was undergoing rapid development, with several companies competing for government contracts. Dissatisfied with his salary and the research facilities at B.F. Goodrich, Wohlgenuth negotiated a new job with International Latex Corporation as manager of engineering for industrial products. International Latex

had just received a large government subcontract for developing the Apollo astronauts' space suits, and that was one of the programs Wohlgemuth would manage. The confidentiality obligation forbid Wohlgemuth from revealing any trade secrets of Goodrich to his new employer. This was easier said than done. Of course it is possible for employees in his situation to refrain from explicitly stating processes, formulas, and material specifications. Yet in exercising their general skills and knowledge, it is virtually inevitable that some unintended "leaks" will occur. An engineer's knowledge base generates an intuitive sense of what designs will or will not work, and trade secrets form part of this knowledge base. To fully protect the secrets of an old employer on a new job would thus virtually require that part of the engineer's brain be removed. Is it perhaps unethical, then, for employees to change jobs in cases where unintentional revelations of confidential information are a possibility? Some companies have contended that it is. Goodrich, for example, charged Wohlgemuth with being unethical in taking the job with International Latex. Goodrich also went to court seeking a restraining order to prevent him from working for International Latex or any other company that developed space suits. The Ohio Court of Appeals refused to issue such an order, although it did issue an injunction prohibiting Wohlgemuth from revealing any Goodrich trade secrets. Their reasoning was that although Goodrich had a right to have trade secrets kept confidential, it had to be balanced against Wohlgemuth's personal right to seek career advancement. And this would seem to be the correct moral verdict as well.

iii. Confidentiality and Management Policies. What might be done to recognize the legitimate personal interests and rights of engineers and other employees while also recognizing the rights of employers in this area?⁴ One approach is to use employment contracts that place special restrictions on future employment. Traditionally, those restrictions centered on the geographical location of future employers, the length of time after leaving the present employer before one can engage in certain kinds of work, and the type of work it is permissible to do for future employers. Thus, Goodrich might have required as a condition of employment that Wohlgemuth sign an agreement that if he sought work elsewhere he would not work on space suit projects for a competitor in the United States for five years after leaving Goodrich. Yet such contracts are hardly agreements between equals, and they threaten the right of individuals to pursue their careers freely. For this reason the courts have tended not to recognize such contracts as binding, although they do uphold contractual agreements forbidding the disclosure of trade secrets. A different type of employment contract is perhaps not so threatening to employee rights in that it offers positive benefits in exchange for the restrictions it places on future employment. Consider a company that normally does not have a portable pension plan. It might offer such a plan to an engineer in exchange for an agreement not to work for a competitor on certain kinds of projects for a certain number of years after leaving the company. Or another clause might offer an employee a special postemployment annual consulting fee for several years on the condition that he or she not work for a direct competitor during that period. Other tactics aside from employment contract provisions have been attempted by various companies. One is to place tighter controls on the internal flow of information by restricting access to trade secrets except where absolutely essential. The drawback to this approach is that it may create an atmosphere of distrust in the workplace. It might also stifle creativity by lessening the knowledge base of engineers involved in research and development. One potential solution is for employers to help generate a sense of professional responsibility among their staff that reaches beyond merely obeying the directives of current employers. Engineers can then develop a real sensitivity to the moral conflicts they may be exposed to by making certain job changes. They can arrive at a greater appreciation of why trade secrets are important in a competitive system, and they can learn to take the steps necessary to protect them. In this way, professional concerns and employee loyalty can become intertwined and reinforce each other.

iv. Confidentiality: Justification: On what moral basis does the confidentiality obligation rest, with its wide scope and obvious importance? The primary justification is to respect the autonomy (freedom, selfdetermination) of individuals and corporations and to recognize their legitimate control over some private information concerning themselves.⁵ Without that control, they could not maintain their privacy and protect their self-interest insofar as it involves privacy.

Just as patients should be allowed to maintain substantial control over personal information, so employers should have some control over the private information about their companies. All the major ethical theories recognize the importance of autonomy, whether it is understood in terms of rights to autonomy, duties to respect autonomy, the utility (as in utilitarian ethics) of protecting autonomy, or the virtue of respect for others. Additional justifications include trustworthiness: Once practices of maintaining confidentiality are established socially, trust and trustworthiness can grow. Thus, when clients go to attorneys or tax accountants they expect them to maintain confidentiality, and the professional indicates that confidentiality will be maintained. Similarly, employees often make promises (in the form of signing contracts) not to divulge certain information considered sensitive by the employer. In addition, there are public benefits in recognizing confidentiality relationships within professional contexts. For example, if patients are to have the best chances of being cured, they must feel completely free to reveal the most personal information about themselves to physicians, and that requires trust that the physician will not divulge private information. Likewise, the economic benefits of competitiveness within a free market are promoted when companies can maintain some degree of confidentiality about their products. Developing new products often requires investing large resources in acquiring new knowledge. The motivation to make those investments might diminish if that knowledge were immediately dispersed to competitors who could then quickly make better products at lesser cost, as they did not have to make comparable investments in research and development. Confidentiality has its limits, particularly when it is invoked to hide misdeeds. Investigations into a wide variety of white-collar crimes covered up by management in industry or public agencies have been thwarted by invoking confidentiality or false claims of secrecy based on national interest. And the possibility of justified whistle-blowing, discussed in Chapter 7, raises additional complications.

v. Conflicts of Interests. We turn now to some equally thorny issues concerning conflicts of interest. Professional conflicts of interest are situations where professionals have an interest that, if pursued, might keep them from meeting their obligations to their employers or clients. Sometimes such an interest involves serving in some other professional role, say, as a consultant for a competitor's company. Other times it is a more personal interest, such as making substantial private investments in a competitor's company. Concern about conflicts of interest largely centers on their potential to distort good judgment in faithfully serving an employer or client.⁶ Exercising good judgment means arriving at beliefs on the basis of expertise and experience, as opposed to merely following simple rules. Thus, we can refine our definition of conflicts of interest by saying that they typically arise when two conditions are met: (1) The professional is in a relationship or role that requires exercising good judgment on behalf of the interests of an employer or client, and (2) the professional has some additional or side interest that could threaten good judgment in serving the interests of the employer or client—either the good judgment of that professional or the judgment of a typical professional in that situation. Why the reference to “a typical professional”? There might be conclusive evidence that the actual persons involved would never allow a side interest to affect their judgment, yet they are still in a conflict of interest. “Conflict of interest” and “conflicting interests” are not synonyms.⁷ A student, for example, may have interests in excelling on four final exams. She knows, however, that there is time to study adequately for only three of them, and so she must choose which interest not to pursue. In this case “conflicting interests” means a person has two or more desires that cannot all be satisfied given the circumstances. But there is no suggestion that it is morally wrong or problematic to try pursuing them all. By contrast, in professional conflicts of interest it is often physically or economically possible to pursue all of the conflicting interests but doing so would be morally problematic. Because of the great variety of possible outside interests, conflicts of interest can arise in innumerable ways, and with many degrees of subtlety. We will sample only a few of the more common situations involving (1) gifts, bribes, and kickbacks, (2) interests in other companies, and (3) insider information.

vi. Gifts, Bribes and Kickbacks. A bribe is a substantial amount of money or goods offered beyond a stated business contract with the aim of winning an advantage in gaining or keeping the contract, and where the advantage is illegal or otherwise unethical.⁸ Substantial is a vague term, but it alludes to amounts, beyond acceptable gratuities, that are sufficient to distort the judgment of a typical person. Typically, although not always, bribes are made in secret. Gifts are not bribes as long as they are small gratuities offered in the normal conduct of business. Prearranged payments made by contractors to companies or their representatives in exchange for contracts actually granted are called kickbacks. When suggested by the granting party to the party bidding on the contract, the latter often defends its participation in such an arrangement as having been subjected to “extortion.” Often, companies give gifts to selected employees of government agencies or partners in trade. Many such gifts are unobjectionable, some are intended as bribes, and still others create conflicts of interest that do not, strictly speaking, involve bribes. What are the differences? In theory, these distinctions may seem clear, but in practice they become blurry. Bribes are illegal or immoral because they are substantial enough to threaten fairness in competitive situations, whereas gratuities are of smaller amounts. Some gratuities play a legitimate role in the normal conduct of business, whereas others can bias judgment like a bribe does. Much depends on the context, and there are many gray areas, which is why companies often develop elaborate guidelines for their employees. What about gifts in routine business contexts? Is it all right to accept the occasional luncheon paid for by vendors giving sales presentations, or a gift one believes is given in friendship rather than for influence? Codes of ethics sometimes take a hard line in forbidding all such gratuities, but many employers set forth more flexible policies. Company policies generally ban any gratuities that have more than nominal value or exceed widely and openly accepted normal, business practice. An additional rule of thumb is: “If the offer or acceptance of a particular gift could have embarrassing consequences for your company if made public, then do not accept the gift.”

vii. Right to Equal Opportunity: Preventing Sexual Harassment: One definition of sexual harassment is: “the unwanted imposition of sexual requirements in the context of a relationship of unequal power.”²⁴ It takes two main forms: quid pro quo and hostile work environment. Quid pro quo includes cases where supervisors require sexual favors as a condition for some employment benefit (a job, promotion, or raise). It can take the form of a sexual threat (of harm) or sexual offer (of a benefit in return for a benefit). Hostile work environment, by contrast, is any sexually oriented aspect of the workplace that threatens employees’ rights to equal opportunity. It includes unwanted sexual proposals, lewd remarks, sexual leering, posting nude photos, and inappropriate physical contact. What is morally objectionable about sexual harassment? Sexual harassment is a particularly invidious form of sex discrimination, involving as it does not only the abuse of gender roles and authority relationships, but the abuse of sexual intimacy itself. Sexual harassment is a display of power and aggression through sexual means. Accordingly, it has appropriately been called

“dominance eroticized.”²⁵ Insofar as it involves coercion, sexual harassment constitutes an infringement of one’s autonomy to make free decisions concerning one’s body. But whether or not coercion and manipulation are used, it is an assault on the victim’s dignity. In abusing sexuality, such harassment degrades people on the basis of a biological and social trait central to their sense of personhood. Thus a duty ethicist would condemn it as violating the duty to treat people with respect, to treat them as having dignity and not merely as means to personal aggrandizement and gratification of one’s sexual and power interests. A rights ethicist would see it as a serious violation of the human right to pursue one’s work free from the pressures, fears, penalties, and insults that typically accompany sexual harassment. And a utilitarian would emphasize the impact it has on the victim’s happiness and selffulfillment, and on women in general. This also applies to men who experience sexual harassment.

Right to Equal Opportunity: Nondiscrimination. Perhaps nothing is more demeaning than to be discounted because of one’s sex, race, skin color, age, or political or religious outlook. These aspects of biological

makeup and basic conviction lie at the heart of self-identity and self-respect. Such discrimination—that is, morally unjustified treatment of people on arbitrary or irrelevant grounds—is especially pernicious within the work environment, for work is itself fundamental to a person’s selfimage. Accordingly, human rights to fair and decent treatment at the workplace and in job training are vitally important. Consider the following two examples: 1. An opening arises for a chemical plant manager. Normally such positions are filled by promotions from within the plant. The best qualified person in terms of training and years of experience is an African American engineer. Management believes, however, that the majority of workers in the plant would be disgruntled by the appointment of a nonwhite manager. They fear lessened employee cooperation and efficiency. They decide to promote and transfer a white engineer from another plant to fill the position. 2. A farm equipment manufacturer has been hit hard by lowered sales caused by a flagging produce economy. Layoffs are inevitable. During several clandestine management meetings, it is decided to use the occasion to “weed out” some of the engineers within 10 years of retirement to avoid payments of unvested pension funds. These examples involve (immoral) discrimination. They also involve the violation of the Civil Rights Act of 1964 and the 1967 Age Discrimination in Employment Act, respectively

Ch: 7. Truth and Truthfulness

Whistle-Blowing: Definition Whistle-blowing occurs when an employee or former employee conveys information about a significant moral problem to someone in a position to take action on the problem, and does so outside approved organizational channels (or against strong pressure). The definition has four main parts. 1. Disclosure: Information is intentionally conveyed outside approved organizational (workplace) channels or in situations where the person conveying it is under pressure from supervisors or others not to do so. 2. Topic: The information concerns what the person believes is a significant moral problem for the organization (or an organization with which the company does business). Examples of significant problems are serious threats to public or employee safety and well-being, criminal behavior, unethical policies or practices, and injustices to workers within the organization. 3. Agent: The person disclosing the information is an employee or former employee, or someone else closely associated with the organization (as distinct, say, from a journalist reporting what the whistle-blower says). 4. Recipient: The information is conveyed to a person or organization that is in a position to act on the problem (as distinct, for example, to telling it to a family member or friend who is in no position to do anything).² The desired response or action might consist in remedying the problem or merely alerting affected parties. Using this definition, we will speak of external whistleblowing when the information is passed outside the organization. Internal whistleblowing occurs when the information is conveyed to someone within the organization (but outside approved channels or against pressures to remain silent). The definition also allows us to distinguish between open whistle-blowing, in which individuals openly reveal their identity as they convey the information, and anonymous whistle-blowing, which involves concealing one’s identity. Some instances are partly open and partly anonymous, such as when individuals acknowledge their identities to a journalist but insist their names be withheld from anyone else. Notice that the definition does not mention the motives involved in the whistle-blowing, and hence it avoids assumptions about whether those motives are good or bad. Nor does it assume that the whistle-blower is correct in believing there is a serious moral problem. In general, it leaves open the question of whether whistle-blowing is justified. BART is a case of both internal and external whistle-blowing that IEEE regarded as desirable and admirable. Let us contrast it with a case in which tragedy occurred because whistle-blowing was not engaged in. In 1974 the first crash of a fully loaded DC-10 jumbo jet occurred over the suburbs of Paris; 346 people were killed, a record for a singleplane crash. It was known in advance that such a crash was bound to occur because of the jet’s defective design.³ The fuselage of the plane was developed by Convair, a subcontractor for McDonnell Douglas.

Two years earlier, Convair's senior engineer directing the project, Dan Applegate, had written a memo to the vice president of the company itemizing the dangers that could result from the design. He accurately detailed several ways the cargo doors could burst open during flight, depressurize the cargo space, and thereby collapse the floor of the passenger cabin above. Because control lines ran along the cabin floor, this would mean a loss of full control over the plane. Applegate recommended redesigning the doors and strengthening the cabin floor. Without such changes, he stated, it was inevitable that some DC10 cargo doors would open in midair, resulting in crashes. In responding to this memo, top management at Convair disputed neither the technical facts cited by Applegate nor his predictions. Company officials maintained, however, that the possible financial liabilities Convair might incur prohibited them from passing on this information to McDonnell Douglas. These liabilities could be severe because the cost of redesign and the delay to make the necessary safety improvements would be very high and would occur at a time when McDonnell Douglas would Truthfulness be placed at a competitive disadvantage.

Moral Guidelines: Under what conditions are engineers justified in whistle-blowing? This really involves two questions: When are they morally permitted, and when are they morally obligated, to do so? In our view, it is permissible to whistle-blow when the following conditions have been met.⁴ Under these conditions there is also an obligation to whistle-blow, although the obligation is *prima facie* and in some situations can be overridden by other moral considerations.

1. The actual or potential harm reported is serious.
2. The harm has been adequately documented.
3. The concerns have been reported to immediate superiors.
4. After not getting satisfaction from immediate superiors, regular channels within the organization have been used to reach up to the highest levels of management.
5. There is reasonable hope that whistle-blowing can help prevent or remedy the harm.

These conditions are not always necessary for permissible and obligatory whistle-blowing, however.⁵ Condition 2 might not be required in situations where cloaks of secrecy are imposed on evidence that, if revealed, could supposedly aid commercial competitors or a nation's adversaries. In such cases it might be very difficult to establish adequate documentation, and the whistleblowing would consist essentially of a request to the proper authorities to carry out an external investigation or to request a court to issue an order for the release of information. Again, conditions 3 and 4 may be inappropriate in some situations, such as when one's supervisors are the main source of the problem or when extreme urgency leaves insufficient time to work through all regular organizational channels. Finally, when whistle-blowing demands great sacrifices, one cannot overlook that personal obligations to family, as well as rights to pursue one's career, militate against whistle-blowing. Where blowing the whistle openly could result not only in the loss of one's job but also in being blacklisted within the profession, the sacrifice may become supererogatory—more than is morally obligatory. Engineers share responsibilities with many others for the products they help create. It seems unfair to demand that one individual bear the harsh penalties for picking up the moral slack for other irresponsible persons involved. Most important, the public also shares some responsibilities for technological ventures and hence for passing reasonable laws protecting responsible whistle-blowers. When those laws do not exist or are not enforced, the public has little basis for demanding that engineers risk their means of livelihood.⁶ Would Applegate have been justified had he decided to whistleblow? As a loyal employee Applegate had a responsibility to follow company directives, at least reasonable ones. Perhaps he also had family responsibilities that made it important for him not to jeopardize his job. Yet as an engineer he was obligated to protect the safety of those who would use or be affected by the products he designed. Given the great public hazard involved, few would question whether it would be morally permissible for him to blow the whistle, either to the FAA or to the newspapers. Was he also morally obligated to blow the whistle? We leave this as a discussion question. Not all whistle-blowing, of course, is admirable, obligatory, or even permissible. Certainly inaccurate whistle-blowing can cause unjustified harm to companies that unfairly receive bad publicity that hurts employees, stockholders, and sometimes

the economy.⁷ But is there a general presumption against whistle-blowing that at most is overridden in extreme situations? Taken together, loyalty, collegiality, and respect for authority do create a presumption against whistle-blowing, but it is a presumption that can be overridden. Loyalty, collegiality, and respect for authority are not excuses or justification for shielding irresponsible conduct. In addition to these corporate virtues, there are public-spirited virtues, especially respect for the public's safety.

Protecting Whistle-Blowers Most whistle-blowers have suffered unhappy fates. In the words of one lawyer who defended a number of them: "Whistleblowing is lonely, unrewarded, and fraught with peril. It entails a substantial risk of retaliation which is difficult and expensive to challenge."⁸ Yet the vital service to the public provided by many whistle-blowers has led increasingly to public awareness of a need to protect them against retaliation by employers. In particular, whistle-blowers played a vital role in informing the public and investigators about recent corporate and government scandals, including whistle-blowers Sherron Watkins at Enron, Cynthia Cooper at WorldCom, and Coleen Rowley at the Federal Bureau of Investigation (FBI)—three women who appeared on the cover of Time Magazine as "Persons of the Year" for 2002.⁹ Government employees have won important protections. Various federal laws related to environmental protection and safety and the Civil Service Reform Act of 1978 protect them against reprisals for lawful disclosures of information believed to show "a violation of any law, rule, or regulation, mismanagement, a gross waste of funds, an abuse of authority, or a substantial and specific danger to public health and safety."¹⁰ In the private sector, employees are covered by statutes forbidding firing or harassing of whistleblowers who report to government regulatory agencies the violations of some 20 federal laws, including those covering coal mine safety, control of water and air pollution, disposal of toxic substances, and occupational safety and health. In a few instances, unions provide further protection.¹¹ Laws, when they are carefully formulated and enforced, provide two types of benefits for the public, in addition to protecting the responsible whistle-blower: episodic and systemic. The episodic benefits help to prevent harm to the public in particular situations. The systemic benefits send a strong message to industry to act responsibly or be subject to public scrutiny once the whistle is blown. Laws alone will usually not suffice, however. When officialdom is not ready to enforce existing laws—or introduce obviously necessary laws—engineering associations and employee groups need to act as watchdogs ready with advice and legal assistance. Successful examples are the Government Accountability Project (GAP) and some professional societies that report the names of corporations found to have taken unjust reprisals against whistle-blowers.

Truthfulness The standard of truthfulness in engineering is very high, much higher than in everyday life. It imposes what many consider an absolute prohibition on deception, and in addition it establishes a high ideal of seeking and speaking the truth. It also generates an array of issues concerning truthfulness in research and academia. In everyday life, the exact requirements of truthfulness are often blurry and contested. Most of us are deeply offended when others deceive (intentionally mislead) us, especially when they lie (intentionally state what they know is false). At the same time, we know of situations that require something less than complete candor, for example to protect our privacy from the intrusive questions of a stranger. Sissela Bok, however, contends that our society has gone too far in creating a climate of dishonesty. She acknowledges the need for occasional lies, for example, to protect innocent lives, and for instances of withholding truths to protect privacy rights. Yet she urges us all to embrace what she calls the principle of veracity: There is a strong presumption against lying and deception, although the presumption can occasionally be overridden by other pressing moral reasons in particular contexts.¹⁸ Even Bok's principle of veracity is too weak to capture the standard of honesty in engineering. Because so much is at stake in terms of human safety, health, and well-being, engineers are required and expected to seek and to speak the truth conscientiously and to avoid all acts of deception. To be sure, confidentiality requirements limit what can be divulged, but there is a stronger presumption against lying than even Bok's principle of veracity. Two of the six Fundamental Canons in the National Society of Professional Engineers (NSPE)

Code of Ethics concern honesty. Canon 3 requires engineers to “Issue public statements only in an objective and truthful manner,” and Canon 5 requires them to “Avoid deceptive acts.” We will refer to these requirements, taken together, as the truthfulness responsibility: Engineers must be objective and truthful and must not engage in deception. All other engineering codes set forth a statement of this responsibility. The truthfulness responsibility enters often into the cases discussed by the NSPE in its Opinions of the Board of Ethical Review. Here are three examples that the board viewed as violating the NSPE Code of Ethics.¹⁹

1. An engineer who is an expert in hydrology and a key associate with a medium-size engineering consulting firm gives the firm her two-week notice, intending to change jobs. The senior engineer-manager at the consulting firm continues to distribute the firm’s brochure, which lists her as an employee of the firm. (Case 90–4)
2. A city advertises a position for a city engineer/public works director, seeking to fill the position before the incumbent director retires to facilitate a smooth transition. The top candidate is selected after an extensive screening process, and on March 10 the engineer agrees to start April 10. By March 15 the engineer begins to express doubts about being able to start on April 10, and after negotiations the deadline is extended to April 24, based on the firm commitment by the engineer to start on that date. On April 23 the engineer says he has decided not to take the position. (Case 89–2)
3. An engineer working in an environmental engineering firm directs a field technician to sample the contents of storage drums on the premises of a client. The technician reports back that the drums most likely contain hazardous waste, and hence require removal according to state and federal regulations. Hoping to advance future business relationships with the client, the engineer merely tells the client the drums contain “questionable material” and recommends their removal, thereby giving the client greater leeway to dispose of the material inexpensively. (Case 92–6) As these examples suggest, the truthfulness responsibility applies widely and rules out all types of deception. Certainly it forbids lying, that is, stating what one knows to be false with the intention of misleading others. It also forbids intentional distortion and exaggeration, withholding relevant information (except for confidential information), claiming undeserved credit, and other misrepresentations designed to deceive. And it includes culpable failures to be objective, such as negligence in failing to investigate relevant information and allowing one’s judgment to be corrupted.

Trustworthiness Exactly why is truthfulness so important, especially within engineering but also in general? One answer centers on respect for autonomy. To deceive other persons is to undermine their autonomy, their ability to guide their own conduct. Deceit is a form of manipulation that undermines their ability to carry out their legitimate pursuits, based on available truths relevant to those pursuits. In particular situations, this can cause additional harm. Deceivers use other people as “mere means” to their own purposes, rather than respecting them as rational beings with desires and needs. This amounts to a kind of assault on a person’s autonomy.²⁰ Most moral theories defend truthfulness along these lines. Duty ethics, for example, provides a straightforward foundation for truthfulness as a form of respect for a person’s autonomy. Rights ethics translates that idea into respect for a person’s rights to exercise autonomy (or liberty). Rule-utilitarianism emphasizes the good consequences that flow from a rule requiring truthfulness. Virtue ethics affirms truthfulness as a fundamental virtue, and it underscores how honesty contributes to desirable forms of character for engineers, the internal good of the social practice of engineering, and the wider community in which that practice is embedded. In addition, each of these ethical theories highlights additional wrongs in how deception harms others in engineering. Dishonest engineering causes financial losses, injuries, and death. Especially important, violating the truthfulness responsibility undermines trust. Honesty has two primary meanings: (1) truthfulness, which centers on meeting responsibilities about truth, and (2) trustworthiness, which centers on meeting responsibilities about trust. The meanings are interwoven because untruthfulness violates trust, and because violations of trust typically involve deception. Engineering, like all professions, is based on exercising expertise within fiduciary (trust) relationships to

provide safe and useful products. Untruthfulness and untrustworthiness undermine expertise by corrupting professional judgments and communications. They also undermine the trust of the public, employers, and others who must rely on engineers' expertise. Sound engineering is honest; dishonesty is bad engineering.

Academic Integrity: Students: Honesty as an engineer begins with honesty in studying to become an engineer. Studies of colleges and universities reveal alarming statistics about academic integrity. According to one study, among schools lacking a strong honor code, three out of four students admitted to having engaged in academic dishonesty at least once during their college career.²¹ Among the schools with an honor code, one in two students made the same admission. Academic dishonesty among students takes several forms.²²

Cheating: intentionally violating the rules of fair play in any academic exercise, for example, by using crib notes or copying from another student during a test.

Fabrication: intentionally falsifying or inventing information, for example, by faking the results of an experiment.

Plagiarism: intentionally or negligently submitting others' work as one's own, for example, by quoting the words of others without using quotation marks and citing the source.

Facilitating academic dishonesty: intentionally helping other students to engage in academic dishonesty, for example, by loaning them your work.

Misrepresentation: intentionally giving false information to an instructor, for example, by lying about why one missed a test. Failure to contribute to a collaborative project: failing to do one's fair share on a joint project.

Sabotage: intentionally preventing others from doing their work, for example, by disrupting their lab experiment.

Theft: stealing, for example, stealing library books or other students' property.

Why do students engage in academic dishonesty?²³ Studies reveal a variety of motivations, including performance worries, such as fear of low grades and desires for higher grades; responses to external pressures, such as heavy workload, parental pressure, or losing financial aid; the belief that professors are unfair, whether in demanding too much or in how they design tests and grades; the desire to help a friend; the belief that because other students are cheating it is all right for me to do the same; and the belief that plagiarism is not a big deal, and that it is a "victimless crime" in which no one really gets hurt. We should also ask why students do not cheat, that is, why they meet standards of academic integrity. Here, too, there are many motives: the conviction that dishonesty is wrong and unfair; the conviction that cheating undermines the meaning of achievement; self-respect; respect for the teacher; and fear of getting caught. Explanation is one thing, justification is another. Is there any justification for cheating, or are proffered justifications simply rationalization—that is, biased and distorted reasoning? As authors, and like most educators, we take a firm stand. Academic dishonesty is a serious offense. It violates fair procedures. It harms other students who do not cheat by creating an undeserved advantage in earning grades. It is untruthful and deceives instructors and other members of an academic community. It violates trust—the trust of professors, other students, and the public who expect universities to maintain integrity. It undermines one's own integrity. And it renders dishonest and hollow any achievement or recognition based on the cheating. Given the seriousness of academic dishonesty, and aware that we are all vulnerable to temptation, what can be done to foster academic integrity?²⁴ Researchers make several recommendations. Universities, as organizations, need to create and maintain a culture of honesty. Honor codes, which set forth firm standards and require students and faculty to report that cheating is going on, make a dramatic difference, even though they are not enough. Especially important, universities must support professors and students who follow university policies in reporting cheating, refusing to bow to the current market mentality in higher education that is more concerned about losing a paying "customer" than about ensuring academic integrity. In addition, professors need to maintain a climate of respect, fairness, and concern for

students. Course requirements and restrictions need to be explained clearly and then implemented. Tests and assignments need to be reasonable in terms of matching the material studied in class, and helpful feedback should be provided as the course progresses. Opportunities to cheat should be minimized. Firm and enforced disciplinary procedures are essential. Just as the Internet has made cheating easier, so has detecting plagiarism been made easier using new Web services. And, we might add, teaching about academic integrity can be a valuable way to integrate an ethics component into courses.

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